

CHAPTER 1

INTRODUCTION

Background of the Study

Horror video games create an enhanced sense of fear compared to film and television of the same genre due to their use of ludic elements. These ludic elements, specifically haptic game design mechanics, play conditioning, and the coding of the games themselves all bypass the screen and controller to impact the body of the player. Understanding affect as the relationship of the body to the world, video games are therefore a strong affective medium for the horror genre. Three aspects enable video games to have this strong impact: haptic game design mechanics, play conditioning, and game coding. Using the controller as a threshold object, video games physically interact with players to create a horror experience that compels the body to mirror the fear on screen. Additionally, horror games condition players to play them in a specific way to heighten tension and deny control. Finally, the coding of horror games can create obstacles that deny player control and create an environment of fear (Hawco, 2023).

In today's world, horror games has been important in the game industry that can be used also in the education sector. Despite their origins as entertainment, horror games have gained recognition for their potential as educational tools. Through the use of emotionally charged, stressful surroundings, these games can help players learn in a variety of subjects especially in the mathematics subject.



In recent years, escape rooms have gained popularity as immersive, interactive experiences that challenge participants to solve puzzles. Some students have disengagement, math anxiety that causes them to have a weak comprehension to solve the math problems. This can affect their learning ability, critical, logical skills and confidence when it comes to solve a mathematical problem.

As the brief history of the students from the department of College of Computer Science and Information Technology, the developers have gathered that students from the College of Computer Science and Information Technology at Golden West College often face difficulties in dealing with their math subjects despite regular instruction and daily practice. Although instructors strive to help them understand mathematical concepts, many students still struggle during activities and examinations. As a result, low test scores are common, which leads to feelings of frustration and discouragement. Over time, this experience contributes to math anxiety, as some students begin to believe they are not capable of understanding or performing well in mathematics.

This study aims to address the challenges faced by students who need to improve their analytical and mathematical skills in the subject of Mathematics through the development and implementation of the game *Escape Room: MathemaTerror*. The main goal of the study is to create an engaging and educational gaming experience that enhances students' problem-solving abilities. Specifically, it seeks to maintain player engagement by combining challenging math puzzles with immersive horror elements, keeping players focused and preventing frustration or fear from causing them to quit.



Additionally, the study aims to develop the *Escape Room: MathemaTerror* game as a tool where students can further enhance their understanding of mathematical concepts. It also ensures that the game is accessible to players with varying learning abilities and skill levels in mathematics. Moreover, the study focuses on designing strategies that motivate players to solve puzzles under pressure while navigating a horror-themed environment. Finally, it emphasizes the importance of maintaining diversity in the puzzles ranging from equations to logic and word problems, to sustain player interest and ensure a balanced progression of difficulty throughout the game.

To address this problem, the developers will be developing a “Escape Room: MathemaTerror” a Math Horror Escape Room, set within a spine-chilling component, participants must use their mathematical knowledge to unravel mysteries, escape traps, and ultimately survive the horrors that lie within. By blending fear with learning, this game aims to increase motivation, improve problem-solving skills, and demonstrate the practical application of mathematical concepts in high-stress, time-sensitive scenarios. The fusion of horror and mathematics not only makes learning more memorable but also encourages collaborative learning through immersive play.



Conceptual Framework

In the development of the game system, the developers use the RAD Model as a software methodology tool. Figure 1.1 shows the paradigm of the study which consist of the input, output, and process.

RAD (Rapid Application Development) model is used in the development of the game entitled “Escape Room: MathemaTerror. The selected methodology incorporates systematic development techniques to the project. This approach prioritizes development and building a prototype rather than planning. With rapid application development, developers can quickly make multiple iterations and updates to the software without starting from scratch. This helps ensure that the final outcome is more quality-focused and aligns with the end users’ requirements.

The process parts of the development of the system are the phases in the RAD (Rapid Application Development) model which includes the Business Modeling, Data Modeling, Process Modeling, Application Generation, and lastly Testing & Turnover.

The software development of the project is based on the Rapid Application Development (RAD) Model. This model focuses on fast and efficient development through continuous feedback and improvement.

After implementing all processes, the output of the study is the creation of Escape Room: MathemaTerror.

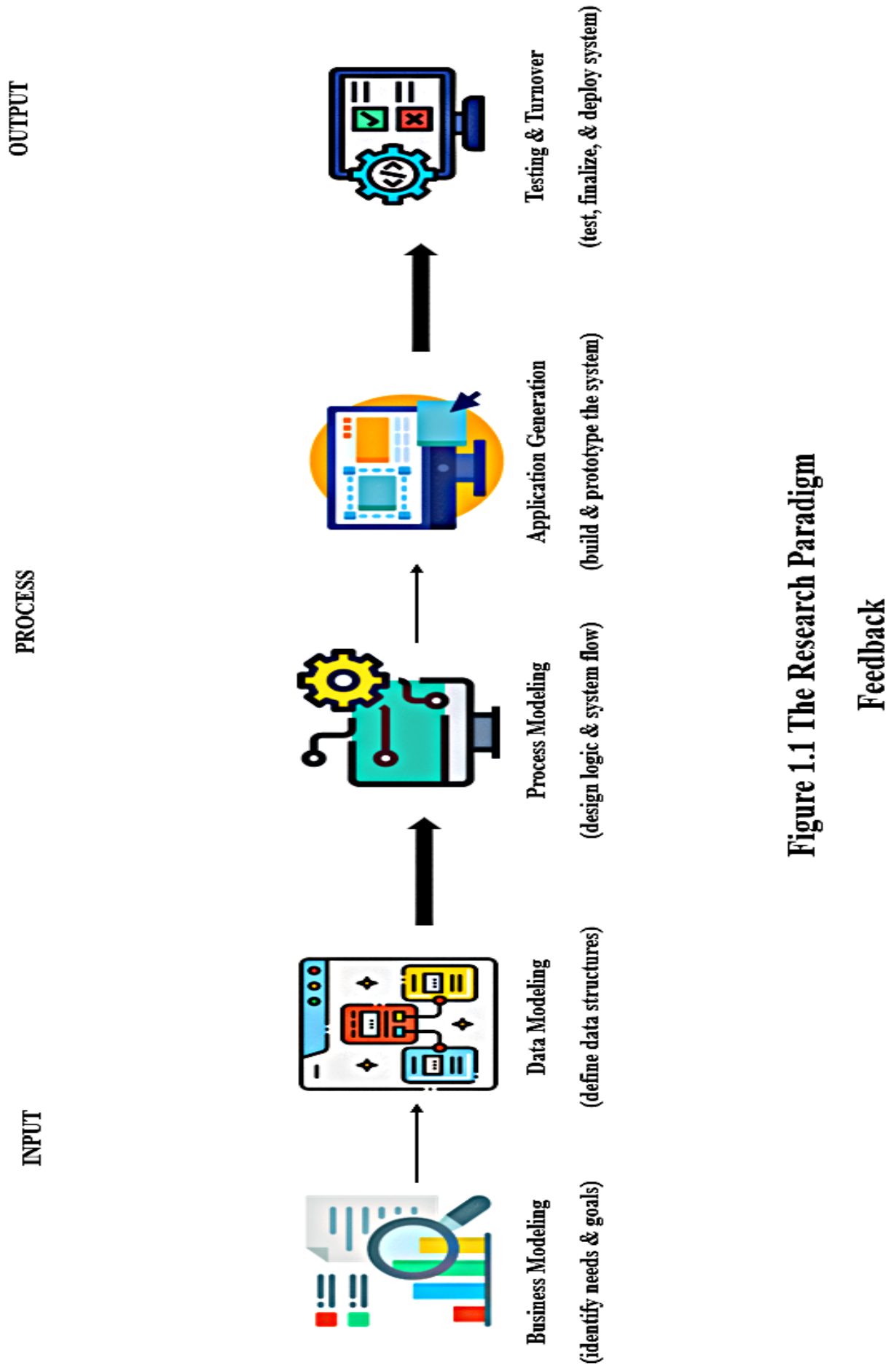


Figure 1.1 The Research Paradigm



Statement of the Objectives

To identify the current processes in the game Escape Room: MathemaTerror, below are the listed to meet the objective of the proposed game for the department of College of Computer Science and Information Technology:

1. To design and utilize a game replicating an escape room to enhance the mathematical skills of the students from the College of Information Technology Education of Golden West Colleges;
2. To describe the features and functionalities of the proposed Escape Room: MathemaTerror game; and
3. To test the usability of the proposed system.

Scope and Delimitation of the Study

This study focuses on developing and designing an “Escape Room: MathemaTerror” game for education and entertainment. This game development, where students will be served as the players of the game in order to enhance their mathematical problem solving skills while experiencing the thrill of playing a horror themed escape room.

This game will feature a first-person perspective where the main gameplay mechanics revolve around solving math-based puzzles to advance through various themed escape rooms, supplemented by time-sensitive challenges to create tension and



urgency during gameplay. Another game mechanism involves consequences for incorrect answers, such as encountering hazardous traps and dangers or facing frightening figures like monsters or ghosts, enhancing the horror-themed escape room experience. For interaction within the game, players can collect items like bandages for healing and pin codes needed to unlock doors, which assist in solving the math puzzles and questions. As for survival mechanics, the health system operates on the premise that incorrect answers will decrease the player's health. Players make three attempts, but if they exhaust their tries without sufficient health, the game will end in a game over. To survive, players must manage their three attempts to progress towards the final challenge of answering questions. Additionally, there will be checkpoints after every five rooms. For example, if a player dies in room 8, they will respawn in room 6. Finally, regarding game progression and difficulty, each room will feature different themes that provide a variety of puzzle challenges, hazards, and repercussions. The game is available in a single version; it can be played either on a computer or a laptop.

The game is designed primarily as a single-player experience; multiplayer or co-op modes may not be available in the initial release. The game contains pre-designed puzzles; players cannot create or customize their own challenges. It also covers basic to advanced math concepts that do not include highly specialized or professional-level



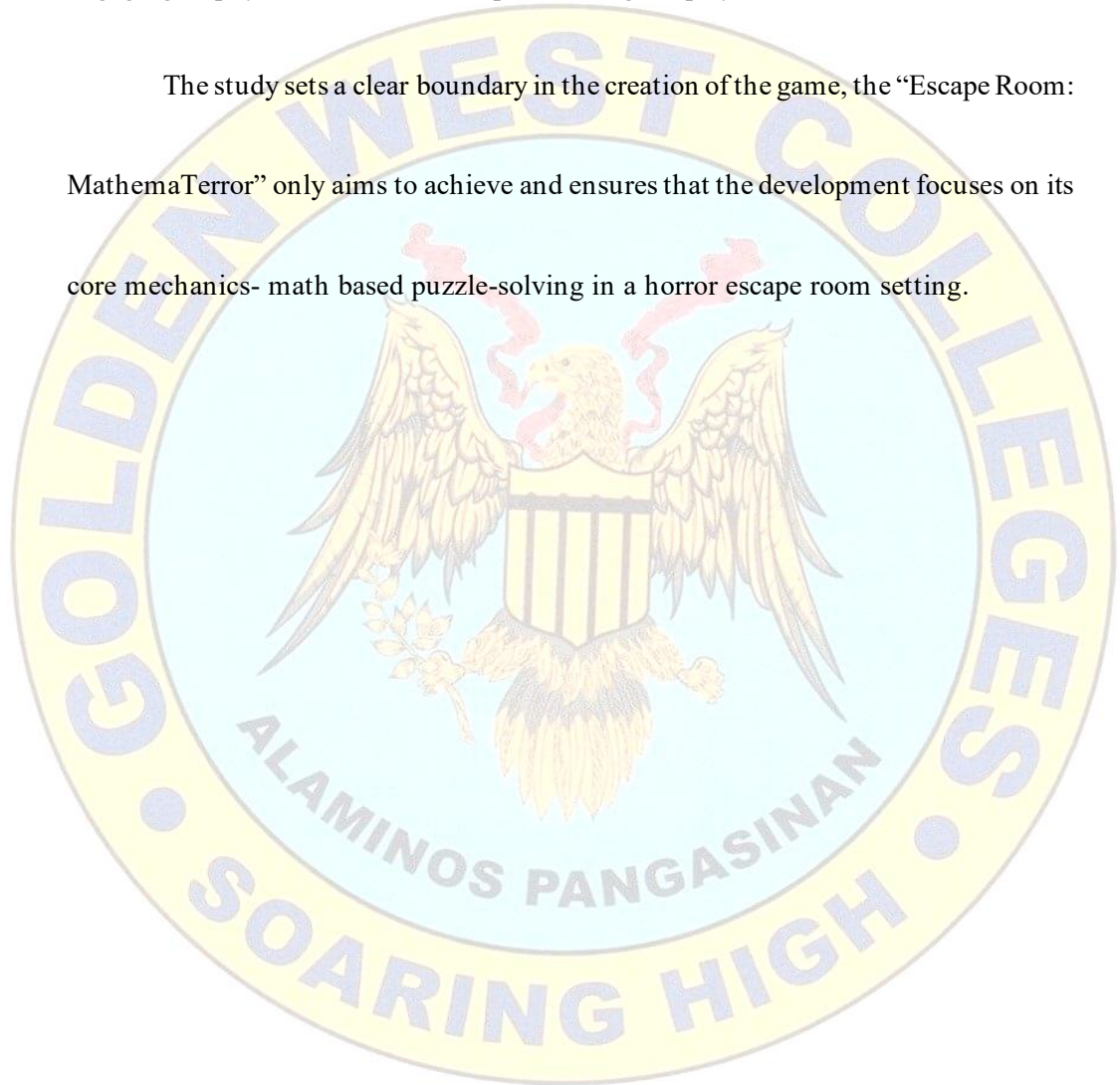
topics. The gameplay only talks about mathematics rather than any other subjects.

Players may also not have the ability to customize their character's appearance or

abilities. At last, players must rely on math-solving skills to escape threats rather than

engaging in physical combat or weapon-based gameplay.

The study sets a clear boundary in the creation of the game, the “Escape Room: MathemaTerror” only aims to achieve and ensures that the development focuses on its core mechanics- math based puzzle-solving in a horror escape room setting.





Significance of the Study

This study is important as it highlights the educational and entertainment value of developing MathemaTerror, a math-based horror escape room game. The findings of this study are expected to contribute to the following:

Golden West Colleges. The game could serve at the institution as a supplement to traditional classroom teaching, allowing students to practice and apply math concepts in a game setting, which could improve their skills and understanding at the institution.

Educators. The game development, Escape Room: MathemaTerror will be useful for teaching math concepts in schools to gain a deeper comprehension of how games might enhance math students' engagement and retention.

Students. The game could benefit from the study, especially those struggling with math anxiety, as it might help reduce fear and promote a more engaging way to practice math.

Developers. This game will broaden the knowledge of the developers on creating the game as well as determine what game development is all about and lastly, what programming can do while developing the game.

Future Developers. This research will be a useful reference for the developers who plan to make any related study exactly the standard underlying the College of Computer Science and Information Technology department.



Definition of Terms

For the purpose of this study, the following terms are defined to establish a clear and consistent understanding of the concepts used. These operational definitions are provided to ensure that readers interpret the terms within the specific context of the proposed Escape Room: MathemaTerror game:

Difficulty progression. The gradual increase in challenge within a game or learning activity to maintain engagement and match player or learner skill levels.

Emerging. New, developing, or growing trends, practices, or technologies in games, education, or research.

Enigmas. Mysterious or puzzling problems that require critical thinking to solve, often used in games or riddles.

Escape rooms. Interactive, themed environments where players solve puzzles and riddles within a time limit to "escape."

First-person point of view. A perspective in games or narratives where players see through the eyes of the character they control.

Foster. To encourage, nurture, or promote the development of a skill, behavior, or mindset.

Horror video games. A game genre designed to evoke fear, suspense, and tension through atmosphere, threats, and narrative.

Immersive. Deep engaging experiences that make players feel present in a game or environment, often through visuals, sound, and interactivity.



Incorporate. To include or integrate elements (such as puzzles, teaching strategies, or mechanics) into a system or activity.

Inclusion. Ensuring that all players or learners, regardless of ability or background, can participate meaningfully.

Ludic elements. Playful features or mechanics in games (e.g., challenges, puzzles, rewards) that enhance engagement.

Pedagogical method. A teaching strategy or approach is used to help students learn effectively.

Retention. The ability of players or learners to remain engaged over time, or the capacity to remember and apply knowledge.

Riddles-filled rooms. Game or learning environments where players must solve puzzles and riddles to progress.

Trap. A hidden or sudden obstacle in games or puzzles designed to hinder progress or increase difficulty.

Unravel mysteries. The process of solving hidden problems, uncovering secrets, or discovering the truth through gameplay or narrative.

Chapter 2

Review of Related Literature and Studies

This chapter provides a detailed review of related literature and studies that serve as the main foundation of this research. It discusses different ideas, findings, and information from previous works that are connected to the topic Escape Room: MathemaTerror. The purpose of this chapter is to give readers a clear understanding of what has already been studied about the subject. It also helps show how this research is related to other studies and how it can contribute new knowledge or insights to the field.

Existing Processes in Escape Room Game

López-Pernas et al. (2019). The study talks about, in addition to being a well-liked form of recreation, escape rooms have drawn the attention of educators due to their ability to foster teamwork, leadership, creative thinking, and communication in a way that is engaging for students. As a consequence, educational escape rooms are emerging as a new type of learning activity under the promise of enhancing students' learning through highly engaging experiences. These activities consist of escape rooms that incorporate course materials within their puzzles in such a way that students are required to master these materials in order to succeed.

Veldkamp et al. (2020), this study about escape rooms (ERs) are team-based games in which players encounter challenges in order to complete a mission in a limited



amount of time. Originally, the nature of the mission was an “escape” from a room. Educators use the game's time constraints and teamwork for different reasons. These educators make different choices for related game aspects such as the structuring of the puzzles. In educational escape rooms more alignment in game and educational aspects is needed.

Morrell & Ball (2019). States that escape rooms, a form of collaborative gaming requiring participants to solve clues to escape a locked room, have grown in popularity across the nation. Games bring together problem solving, critical thinking, and teamwork. This game method has not yet appeared in nursing education literature. The authors describe two versions of the innovative escape room pedagogical method used in undergraduate nursing education. Theoretical bases, learning activities, and types of game clues that could be used are detailed.

Quek et al. (2023), states that educational escape rooms (EERs) are enjoyable and increase learning motivation. Virtual EERs are emerging through use of immersive virtual technology. Escape rooms were employed as learning and assessment strategies for the delivery of diverse topics within dentistry, medical, nursing, pharmacy and physiotherapy pre-registration healthcare education. Most were conducted on nursing students with learning topics focusing on the development of knowledge related to patient care management. Virtual escape rooms became more prevalent following the COVID-19 pandemic. The specific design features of escape rooms were



based on the elements of game-based learning including team learning with appropriate size, time limit and sequential path-design of puzzles.

Botturi and Babazadeh (2020), this study about escape games have become popular nowadays. Groups of people are locked inside riddles-filled rooms and have to work their way out finding clues and solving puzzles. While being fun for groups or team building, escape games have a great educational potential, which can be capitalized if teachers integrate them in their professional activities. In this paper we introduce the Star Model, a tool for teachers to design and play their own educational escape rooms, which was built on top of previous literature and experiences.

Pato et al. (2024). The study talks about an escape room, which is a game in which a player or team of players is locked in a room or series of rooms. To exit the room successfully, they must solve puzzles or search for clues within a set amount of time. While commercial escape rooms are typically designed for entertainment, an escape room format can be used as an educational experience as well. A well-designed escape room can heighten engagement, provide immediate feedback, and help practice and cooperation. This paper describes the design and implementation of an educational escape rooms class at a major university in the Philippines. It describes the class's goals and objectives, and the major elements of an escape room's design.

Clarke et al. (2017). This study talks about game-based learning (GBL) is often found to be technologically driven and more often than not, serious games for instance, are conceptualized and designed solely for digital platforms and state of the art



technologies. To encourage a greater discussion on the potential benefits and challenges of a more holistic approach to developing GBL that promote human centered interactions and play for learning, the authors present the escapED program. The escapED program was conceived following the recent entertainment trend of escape rooms and is used for developing non-digital GBL approaches within education. escapED aids the design and creation of educational Escape Rooms and Interactive Gaming Experiences for staff and students in further/higher education settings.

Videnovik et al. (2022). A study about starting from students' perspective, they created a simple digital escape room-style game where students got an opportunity to self-assess their knowledge in computer science at their own pace. Students tested this prototype game, and their opinions about the game were collected through an online survey. The test's goal was to evaluate the students' perceptions about the implemented digital escape room-style educational game and gather information about whether it could achieve students' engagement in learning computer science during online teaching.

Lathwesen & Belova (2021). A study about an integral part of escape rooms are puzzles. In commercial escape rooms, a distinction is made between two different types of puzzles: mental and physical puzzles. To solve the first, clues must be discovered, deciphered, and related to each other. This requires cognitive skills and logical thinking. The counterpart to this are physical puzzles or tasks in which real objects or parts of the room have to be moved to find the solution. They are more time-



consuming and represent cognitive relief. Both types of puzzles are also used in combination. Despite the variety of possible puzzles, three basic structural components can be identified: a problem, a hidden solution, and a reward. To receive the reward, players must first decipher the puzzle and complete the resulting challenge. Often the solution is already hidden in the puzzle itself. The reward may include new puzzle pieces, clues or objects.

Vankúš (2021). In modern education nowadays, the use of game-based learning as a teaching and learning method is popular in all school subjects, including mathematics. There are numerous studies dealing with the influences of this teaching method on the students' achievements. Modern teaching theories consider an important effect of education to be on the development of students' affective domain, connected with the subject and its teaching.

The existing process in developing and implementing escape room games ranges from traditional, manually operated setups to technologically enhanced or fully digital systems. Similar to how automation in other systems promotes transparency and efficiency, the integration of digital tools in escape rooms enhances engagement, interactivity, and learning outcomes.

Features of Escape Room Game

González-Yubero et al. (2023). In order to motivate students towards learning, gamification through the escape room emerges as one of the most innovative techniques in the current educational context. Educational escape rooms are games



whose objective is that several teams of students managing to remove themselves from an enclosed space in which a disturbing event is going to happen. Each group of students must cooperate to solve the riddles that lead to the codes or keys that will unlock the door within a time given by the teacher. The riddles to be overcome can be based on logic tests, search tests, riddles and puzzles, mathematics, etc.

Escaplive (2025). A study about escape rooms offer a dynamic and engaging way to develop a variety of educational skills. By enhancing problem-solving abilities, fostering teamwork, improving communication, and encouraging creative thinking, escape rooms provide a fun and effective platform for learning. Whether used in an educational setting or as a recreational activity, the benefits of escape rooms extend far beyond entertainment, making them a valuable tool for personal and professional development.

Dugnol-Menéndez et al. (2021). This is a study of escape rooms are described as “live-action team-based games where players discover clues, solve puzzles, and accomplish tasks in one or more rooms, in order to accomplish a specific goal, usually escaping from the room, in a limited amount of time”. This offers a non-traditional, experiential, peer-group learning opportunity that fosters constructive interactions, leading to observations of one’s own and others’ unique leadership skills and styles.

Çakmak and Kaymaz (2024). When describing escape room games as a gamification methodology, key features include the presence of a problem or challenge within the gamification content, the process being guided by instructions, and the



incorporation of an enjoyable learning experience . Since escape room games require problem-solving, critical thinking, and active participation in decision-making within a team, they engage an individual's cognitive processes throughout the gameplay. The challenges and puzzles integrated into the game content encourage players to think creatively and strategically, enhancing their problem-solving skills.

Sarin et al. (2025). This study talks about an ERs, which are a form of educational tool using games and some activities to promote active learning. Learners can retain the information for a long time with skill. This approach has been adopted in education and is widely used in different fields. In the ERs, students solve the challenges within a limited time to achieve their goals. Primarily based on escaping a locked room, modern technique includes the tasks like investigating the crime scene and murder mystery or breaking into a vault. These ERs are developed by passionate educators and based on their experience with recreational ERs or video games which shared materials on various platforms.

Pornsakulpaisal et al. (2023). A study about design teams must also create puzzles, another core game element for DEERs. At its most basic structural level, a DEER consists of several puzzles, with players using clues (keys), to solve puzzles (locks), to 'escape' the game. In the online setting, puzzles tend to include codes, crosswords, riddles and maths or pattern-based puzzles. Design teams can develop their own 'home-made' puzzles or use one of the many freely available online puzzle-building websites (e.g., Jigsaw Planet, Breakout EDU).



Albrecht et al. (2025). This study talks about educational escape rooms (eER) offer great potential to reach this target group. Educational escape rooms are defined as pedagogical activities that are time-constrained and problem-based and offer a good opportunity to address problem-oriented learning by developing tasks that are based on an authentic problem and embedded in a meaningful scenario. Key features of an escape room activity are solving puzzles under time constraints in the context of a storytelling narrative.

Helbing et al. (2023). A study about educational escape rooms operate in generally the same manner but are developed for a specific audience with well-defined learning objectives. From an instructional standpoint, escape room games are collaborative, task-centered, time-based, and provide immediate feedback, making them an attractive consideration for in-class learning activities. In developing escape room activities for optometry, pharmacy, and medical students, the prompt for each escape room scenario was formatted as a research proposal. The rationale was to connect the development of library skills to a real-world, practical, and beneficial outcome.

Lim (2023). Employing educational escape rooms is a novel game-based learning strategy in various disciplines, including health sciences. Escape rooms provide the opportunity for a group to work together as they solve puzzles within a limited time. The skills harnessed and developed within the game by participants are parallel to these soft skills.



Mathieson and Duca (2021). Escape rooms are themed rooms in which players are locked inside for a specific amount of time, usually one hour. The players must solve several puzzles in order to find a key and escape from the room before the time limit. Variations now exist where the primary goal is to solve a mystery or unlock an item. Puzzles are highly varied, from logical to abstract, from physical to mental, from individual to team based. They will often build from each other and use props and technology.

The features of escape room games have evolved from simple physical challenges to complex, technology-enhanced experiences designed to promote learning, teamwork, and problem-solving.

Usability of Escape Room Game

Ang et al. (2020). An escape room uses a game-based active learning approach to develop students' critical thinking and problem-solving skills in an immersive environment. An escape room was developed to provide an innovative method to reinforce concepts in chemical bonding for a first-year general chemistry course. Because of the COVID-19 pandemic, it was unsuitable to implement the escape room in a physical setting, as students need to interact in close proximity. To circumvent this problem, a fully digital escape room was developed, as it is cost-effective, accessible, and easy to use. Students' feedback indicates that both escape rooms reinforce and motivate learning.



Makri et al. (2021). Educational Escape Rooms (EERs) with the use of digital technologies. More specifically, the focus of the study is to present the current developments and trends concerning Digital Educational Escape Rooms (DEERs) and investigate how they foster learning outcomes for online learners. Additionally, the present study provides insights into the design process of such technology enhanced EERs. Both creating and using ER breakout activities can be deemed as instructional strategies that can enliven classroom learning experiences and lead to more learner-focused strategies. They are learner-centered activities in which a team of players cooperatively discover clues, solve puzzles, and complete tasks to progress through the challenge to achieve a specific goal. Such game-based methodologies are rapidly increasing in terms of their applicability, especially in higher educational institutions

Fotaris and Mastoras (2022), escape rooms (ERs) are emerging as a new type of learner-centred activity designed to enhance students' learning and 21st century skills in primary, secondary, higher education, and professional development programs. An educational escape room (EER) can be defined as an instructional method requiring learners to participate in collaborative playful activities explicitly designed for domain knowledge acquisition, skill development, or behavioral change so that they can accomplish a specific goal (e.g., participants must escape from a physical or virtual room, solve a mystery, find a hidden item, prevent a disaster, break into a vault, etc.) by solving puzzles linked to unambiguous learning objectives in a limited amount of time



Borrás-Gené et al. (2022). Teachers often need to adapt their teaching methodologies in order to overcome possible limitations and ensure that education does not lose quality in the face of different scenarios that may arise in the educational environment, which are not always the most desirable. Techniques such as the Educational Escape Room (ERE) in higher education are taking a great increase due to its popularity among young people as a leisure activity. Educational Escape Rooms (EER) are immersive, engaging, dynamic, and activity-oriented online learning experiences that are developed due to their cost-effectiveness, accessibility, and practicality. They possess numerous benefits; among others, they achieve engagement with a learning environment. Furthermore, interaction through collaboration helps learners develop social skills.

La Torre et al. (2024), escape room games are educational gamification technologies that consist of introducing a team of players into a physical or digital space in search of clues to answer puzzles, riddles or enigmas and solve a mystery or problem. The use of educational technologies aims to facilitate and improve learning through the creation, use and management of appropriate technological processes and resources. ER games consist of introducing a team of players in a physical or digital space in search of clues to complete puzzles, riddles or enigmas, with the aim of solving a mystery or a problem. ER games have the aim of acquiring professional skills in a complementary way to other teaching methods. ER games have been used in recent



years in the field of health sciences education, including nursing studies, whether undergraduate or graduate

Bangcaya et al. (2021), educational games may engage learners from being alienated by conventional methods of classroom instruction and have the potential to optimize learning, but motivation or appropriate utilization of gamification design concepts can often be a problem in finding the purpose of the learning activity among researchers and practitioners. This study aimed to examine the learning outcomes and level of motivation of learners using gamification approach in teaching Science.

Boldadora, (2018) The integration of technology continues to increase as the educational system is challenged amidst the endeavors expressed in the traditional philosophy and conventional approaches. While there are many opportunities for the use of different technologies in classroom instructions, there is a need to better understand how effective the digital game-based learning is when integrated into classroom instruction. This study investigated the effectiveness of digital game-based learning on mathematics achievement of students. The respondents of the study were the two sections of grade six students in one of the schools in Davao City. Each section is composed of 44 students. The experimental group received digital game-based learning as innovative approach while the control group was taught using the conventional approach.

Orbon & Sapin (2022), the researchers developed the students' capacity to think, comprehend underlying ideas, and solve challenging mathematical problems



when they learn mathematics through games. While having fun, educational games encourage students to come up with original ideas and push them to learn more quickly. Together, games help learners feel powerful and on par with their peers. Game-based learning using different game application tools emphasizes the importance of strengthening the foundation of math concepts and the mathematical proficiency to solve various problems in math.

Nob et al. (2024). There is a dearth of research investigating the synthesis of the overall effects of gamification in mathematics education within the Philippine context. Thematic analysis revealed that the effects of gamification were multifaceted in the Philippine setting. Positive outcomes include increased engagement and motivation, improved academic performance, and enhanced higher-order thinking skills. Several factors support the increase in student engagement and motivation including the provision of a competitive learning environment, the positive impact of collaboration, the use of rewards, and the role of interactive and immersive features in gamified interventions. Moreover, the improvement in the academic performance of students in math is supported by factors such as the integration of feedback mechanisms, gamifications' ability to meet students' diverse learning needs, and the important role of technology as an enabler of gamified interventions.

Bierbooms et al. (2024). The benefit of an escape room may be that it offers the option to train skills by gaining hands-on experiences (experiential learning). Escape rooms create realistic learning environments that foster meaningful



participation and provide a supportive atmosphere for experimentation. In contrast, other methods (eg, classical instructions or eLearning) mainly focus on providing instructions on using an artifact or service. In addition, an escape room is focused on the learning process while simultaneously being educational and fun, which makes it much more engaging to participate in training.

The usability of escape room games lies in their ability to seamlessly integrate learning, engagement, and technology into a single immersive experience that promotes active participation and deeper understanding. These games are designed not only for entertainment but also as innovative educational tools that encourage learners to apply critical thinking, collaboration, and problem-solving skills in real time. Through the combination of interactive puzzles, time constraints, and engaging narratives, escape rooms transform traditional learning into an experiential process where students learn by doing rather than by passive observation.

This chapter presented a review of studies about the development, features, and usability of escape room games. The findings show that escape rooms have evolved from manual to digital systems that promote teamwork, problem-solving, and engagement through interactive puzzles and time-based challenges. These games effectively combine learning, technology, and fun to create meaningful educational experiences. The current study builds on these ideas by developing an educational escape room system designed to make learning more interactive and effective for students at the College of Computer Science and Information Technology department

Chapter 3

Design and Methodology

This chapter gives an outline of research methods that were followed in the study. It provides information on the participants, that is, the criteria for inclusion in the study, who the participants are and how they will be sampled. The developers describes the research design that was chosen for the purpose of this study and the reasons for this choice. The instrument that will be used for data collection is also described and the procedures that will be followed in carrying out this study are included. The developers also discusses the methods that will be used to analyze the data that will be gathered.

Research Design

Quantitative data provides measurable information that can be used to make decisions, evaluate performance, and analyze trends. Quantitative research is a method of examining data that incorporates statistical techniques. Through usability test survey, researchers can gather quantitative data for study. Based on the patterns found in the data, researchers can use statistical approaches to derive significant findings and make well-informed decisions.

This chapter's objective is to develop the methodology of the study methodology using quantitative research methods. Additionally, the research approach helps the researcher discover the findings of the study. Within this chapter, the overall research design and data collection techniques collection are thoroughly described.



This section's objective is to demonstrate the manner in which the research was carried out during the study period.

The researchers conducted usability test survey research for the students at the College of Information Technology Education Department at a higher education institute like in the Golden West Colleges. The researchers have managed to get information from the students and from this survey the developers gained more information on what features are needed for the system.

Software Development Model

Rapid Application Development, or RAD, is an adaptive software development model based on prototyping and quick feedback with less emphasis on specific planning. In general, the RAD approach prioritizes development and building a prototype rather than planning. With rapid application development, developers can quickly make multiple iterations and updates to the software without starting from scratch. This helps ensure that the final outcome is more quality-focused and aligns with the end users' requirements. Rapid Application Development (RAD) is a highly collaborative model, designed to bring together IT teams, business stakeholders, and end users. This collaboration is essential for ensuring alignment and delivering the best possible product (Kissflow, 2024).



The developers have used the Rapid Application Development also known by its short term “RAD”. In order to speed up the software development of the game which is the “Escape Room: MathemaTerror” as well as highlight quick delivery and user feedback when being tested by the students at the College of Information Technology Education Department. Lastly this is used in order the developers to update and improve the prototype during the development process of the system.

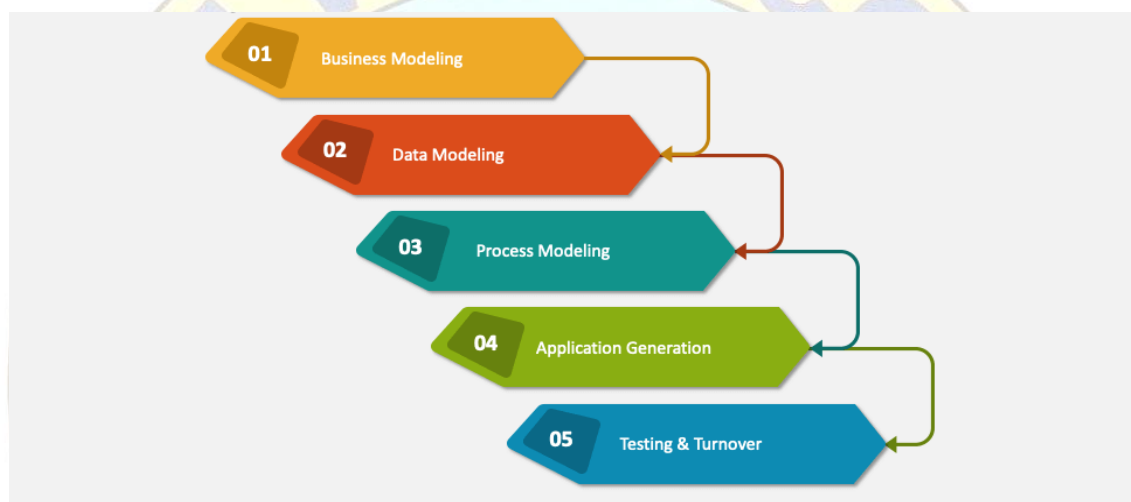


Figure 3.1 SD Model Diagram- RAD (Rapid Application Development)

The Stages of Rapid Application Development

Business Modeling. The first stage of the RAD model is Business Modeling, which aims to understand the key objectives and flow of information within an organization or system. This phase focuses on analyzing what data is vital to the business process, how that data is gathered, and who is responsible for its use. It is where developers and stakeholders collaborate to define the system’s goals, scope, and



user needs. In this study, the developers of Escape Room: MathemaTerror conducted an analysis of the educational environment of the College of Information Technology Education Department at Golden West Colleges., The team gathered essential data about students' challenges and learning patterns in mathematics. The main purpose was to identify the problem areas and determine how a game-based system could enhance mathematical engagement and comprehension. This stage also included defining the target users, setting the system objectives, and outlining the expected outcomes of the game which is an educational horror-themed escape room designed to strengthen students' mathematical problem-solving skills through interactive gameplay.

Data Modeling. The Data Modeling stage involves analyzing and structuring the information collected during business modeling into a framework that supports the system's functionality. It identifies the data objects required, their relationships, and how they can be organized to form a stable foundation for the system's operation. This ensures that the data flow within the system is coherent and efficient. In the context of Escape Room: MathemaTerror, this stage focused on defining the various data elements and their connections. This phase was crucial to guarantee that the educational content aligned with the intended learning objectives, providing a foundation for meaningful gameplay interaction that the players will be playing.

Process Modeling. Once the data model was established, the project moved on to the Process Modeling stage, where the focus shifted to defining how the data is



processed to fulfill the system's goals. This stage translates the data model into detailed procedures that outline how the system behaves and how users interact with it. For Escape Room: MathemaTerror, this meant designing the sequence of events, puzzles, and challenges that the player experiences during gameplay. The developers defined how mathematical problems are presented to the players as well as how the answers will be validated in order to escape a certain room and proceed into the next puzzled room. This stage also included planning the user interface flow, ensuring that the system was easy to navigate and engage for students.

Application Generation. The Application Generation stage is where the actual development and construction of the system take place. It involves converting the designs and models from the previous stages into a working prototype or system. Developers use rapid development tools and programming techniques to build, test, and refine the application quickly. In the RAD model, this phase is all about prototypes that are created, tested by users, and improved multiple times based on feedback. During this phase of Escape Room: MathemaTerror, the development team began coding the initial prototypes of the game. Afterwards each round of feedback was used to refine and improve the system until a stable and well-balanced version of the game was achieved

Testing and Turnover. The last stage of the RAD model is Testing and Turnover. In this stage, the system is carefully tested to make sure it works well and is ready to be used. Developers look for any mistakes, bugs, or problems that could affect



how the game system works and fix them. They also check if the game system meets all the user's needs and works properly in real situations. For Escape Room: MathemaTerror, this phase included extensive testing of all features in the game system. The developers performed several rounds of debugging to ensure that the game was stable, user-friendly, and free of technical issues. Students and instructors participated in playtesting to validate the game's educational effectiveness and engagement value. After ensuring that the system met both technical and educational standards, the developers finalized the documentation and prepared the game for deployment.

Sources of Data

In this study, the developers have collected information from these two sources. Implementing a survey for all the students of the department of College of Information Technology Education Department inside Golden West Colleges wherein it determined whether the students have tried playing an educational learning game, what are their experiences while playing that certain type of game. Another one is what their preferences would be in the creation of the game, which is the Escape Room: MathemaTerror.

Gathering data from the internet is the second and last source of information.

The World Wide Web is a huge data source that the developers used to get information.

The developers gained information like online-related literature, articles, books, and



study papers, which is accessible to anyone, which helps developers find related studies regarding game-based learning to help the developers to develop the system.

All the information collected from these two sources is carefully analyzed and is related to the proposed game, Escape Room: MathemaTerror.

Instrumentation and Data Collection

This section covers the developers data gathering process, which includes a research component to assess and analyze the accuracy of the information for the study.

Survey. It is a way of asking people questions to gather information or opinions about a specific topic. It helps developers gather information about opinions, behaviors, or experiences. Surveys are often used in scientific studies to understand trends and patterns in a population. They can be designed with multiple-choice, yes/no, or open-ended questions. Researchers use the data from surveys to analyze and draw conclusions for their studies. Golden West Colleges' students are being surveyed by the developers in order to obtain feedback and ideas. To find out the brief information from the students that belongs to the College of Information Technology Education Department at Golden West Colleges about their mathematical skills and can be improved by playing the game, Escape Room: MathemaTerror after the development of the game. Simply the developers used surveys to gather data from 30 random people to help the developers understand what the masses want to see in the end product.

Online Research. Online research is a research process that includes getting information from the internet, traditional methods pen-and-paper research methods



have been replaced by online ones since the creation of the internet. Online research methods help people find the information they need quickly. Online tools provide information instantly. This information would take much longer to locate using offline research methods. The developers often look for new material online to speed up the collection of data from different sections related to their research. Developers used the internet to search thoroughly the web to find necessary information to help with the development of the system and act as a guide for the developer.

Tools for Data Analysis

In order to gather information for the planned system's development, the developers employed the following instruments. These gathered data are thoroughly examined in order to comprehend the relevant information for creating the game, Escape Room: MathemaTerror.

The benefits of using a tool for data analysis improve the accuracy, increase efficiency, scalability, accessed advanced analytics, cost saving, and collaboration sharing.

Flowchart. A flowchart is a visual representation of a process, system, or algorithm that illustrates the sequence of steps, decisions, and actions taken to achieve a specific goal or solve a problem. For managing and analyzing the workflow, the developers created a use of flow charts. It is a graphical tool that uses symbols, arrows, and text to depict the flow of logic, making it easier to understand, analyze, and communicate complex processes.



Gantt chart. A Gantt chart is a type of bar chart that illustrates a project schedule, showing the tasks, dependencies, and timelines involved in completing a project. It is a visual representation of a project's workflow, helping to plan, coordinate, and track progress over time. The developers used the Gantt chart to represent tasks or events against times. It is a calendar-like representation of a project's workflow, allowing teams to see the entire project timeline, dependencies, and deadlines in one glance.

Use Case Diagram. A use case diagram is a graphical representation of a system's functional requirements, showing how users interact with the system to achieve specific goals. It is a powerful tool in software development, business analysis, and system design, helping to identify, organize, and prioritize system requirements. The Use Case Diagrams that the developers explained on how the system will function in various scenarios. It is a window into the system's functional soul, providing a deep understanding of the users' needs, goals, and motivations.

Weighted Mean. The weighted mean is a specific type of average that assigns different levels of importance or weight to different values. To calculate it, you multiply the weight of each value by its corresponding numerical result. Afterward, you add up all the resulting products to obtain the final weighted mean. This method ensures that some values contribute more significantly to the average than others, depending on their assigned weight. The developers used this method in order to get more accurate decisions based on different data points that have varying levels of importance.



Requirements Analysis

The Golden West Colleges is an institution from the Western, Pangasinan. It is located in San Jose Drive, Poblacion Alaminos City, Pangasinan. Landmark: Infront of McDonald's. The game will be played by the students at the College of Information Technology Education Department in the institution.



Figure 3.2 Student player accessing the game with their phones

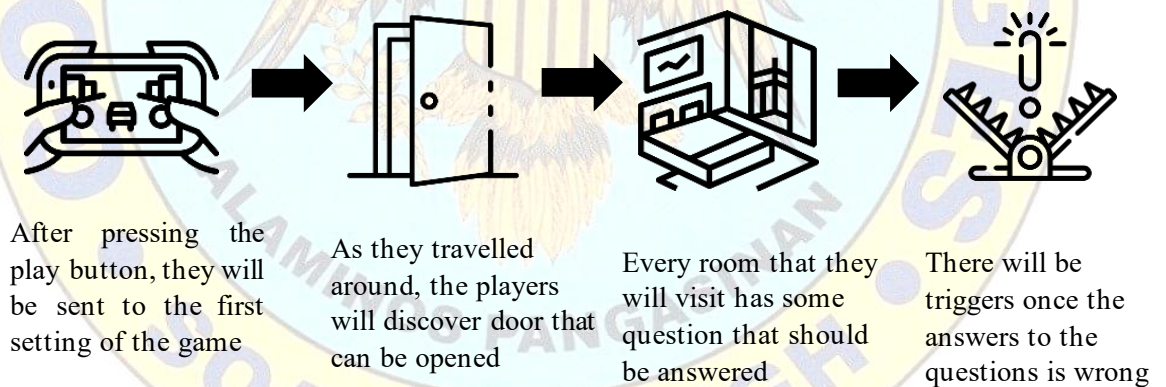
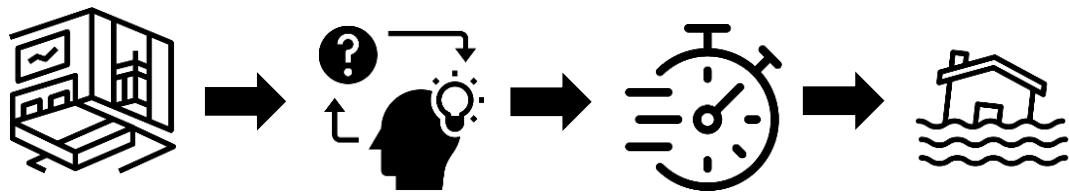


Figure 3.3 Players once its journey in the gameplay starts

To begin with, this is the manual process in figure 3.2 and 3.3. When a player student starts to open the game app on their phones they will be greeted with the home screen display of the game and once they started to play the game they can finally travel the area, as they reached the doors of each room. There will be triggers or traps that players will experience if their answers are wrong.



Every room will have mathematical questions that should be answered

As players entered the room and finally saw the questions

The timer will start moving and they must answer it before it ends

If they did not get the right answer before or within the time, water level will rise inside the room

Figure 3.4 The room will trigger the flood chamber when every answer is wrong

In this figure, just like in figure 3.3 as the players started to answer the question that they will discover in every room they enter, the timer will start running and as time passes by if they got an incorrect answer the flood chamber will trigger wherein the water level will rise inside the room and possible their life bar will decrease but if they successfully got the right answer then the hazard will not be triggered.



Every room will have mathematical questions that should be answered

As players entered the room and finally saw the questions

The timer will start moving and they must answer it before it ends

If they did not get the right answer before or within the time, if not the flying spike trap will be triggered

Figure 3.5 The room will trigger the spike trap when every answer is wrong



In figure 3.5 as the players started to answer the question that they will discover in every room they enter, the timer will start running and as time passes by if they got an incorrect answer the flying spike trap will trigger wherein it can decrease the life bar of the players but if they successfully got the right answer then the hazard will not be triggered. So as much as possible the players must get their answers correctly to prevent the hazards or traps that they may experience while playing.

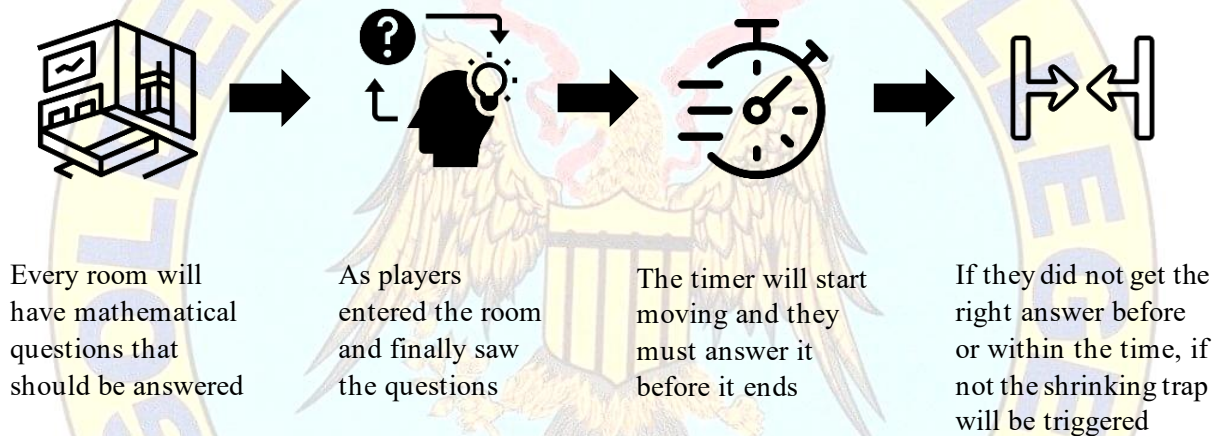
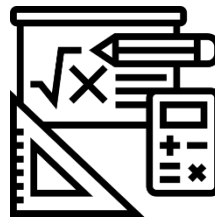


Figure 3.6 The room will trigger the shrinking trap when every answer is wrong

In figure 3.6 as the players started to answer the question that they will discover in every room they enter, the timer will start running and as time passes by if they got an incorrect answer the shrinking trap will trigger wherein it shrinks the room and possible the player cannot escape but if they successfully got the right answer then the trap will not be triggered.



Player travelling and saw the questions



As they start or before answering the questions they will be distracted



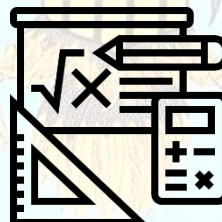
Ghost will distract them whether the players are answering or just travelling the area

Figure 3.7 Ghosts will distract the player whether they are solving or not

When a player starts or before answering the questions that is shown on a room, they will be experiencing chills since there will be jump scares of ghost trolling on them to get them distracted. The ghost will appear not only appear at the players while answering but they will also appear sometimes during the whole game play to make the game more fun and scary.



Player travelling and saw the questions



As they start or before answering the questions they will be distracted



Monsters will distract them whether the players are answering or just travelling the area

Figure 3.8 Monsters will distract the players whether they are solving or not

Just like in figure 3.7 instead of ghost there will be monsters also who will not only appear at the players while or before answering but they will also appear sometimes during the whole game play to make the game more fun and scary.

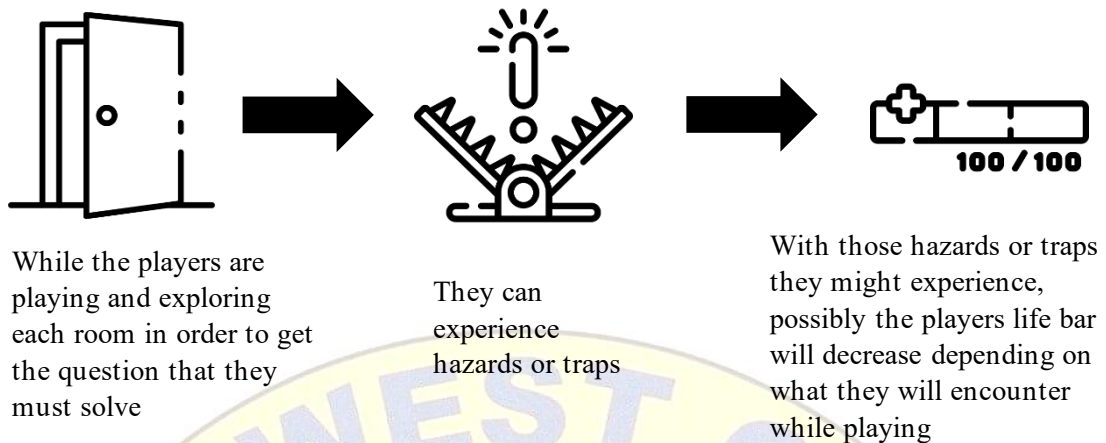


Figure 3.9 Traps and hazards that can decrease players life bar but depending on what they deal with

In this figure, traps or hazards can be seen and experienced by every player who are going to play the game but depending on what they are going to deal with as they explore and try to reach the end in order to survive the whole game play.

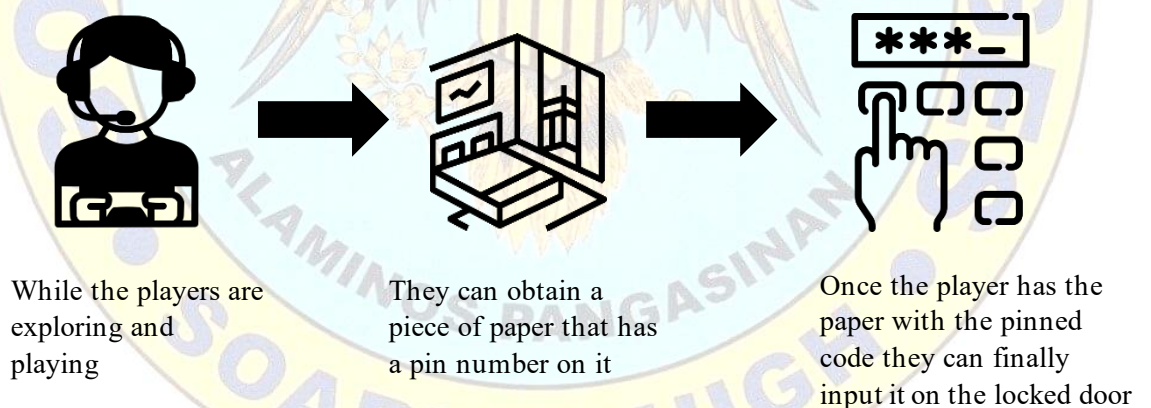
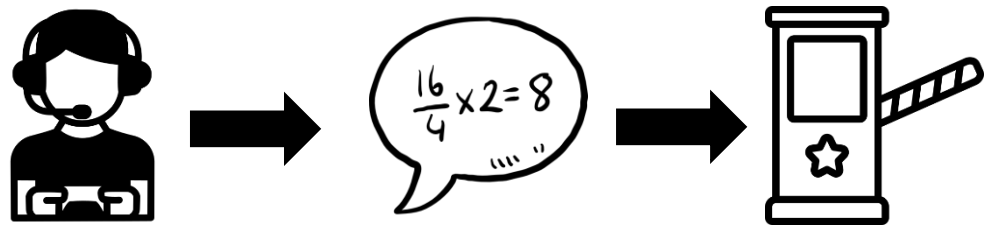


Figure 3.10 Pin numbers on a paper that can be used in locked rooms by entering it on the pinned door

In this figure, as the players travels and roams around a certain area that has a locked room, they can obtain a piece of paper that has a pin number on it wherein this can be used for locked rooms that will be unlocked once the player entered the pin number.



While answering a question that is being shown in the question wall panel.

When player did not get the correct answer on a specific room that has a question that is needed to be answered.

There will be checkpoints in the game that will serve as the player's progress in specific point.

Figure 3.11 Checkpoints as player's progress point

In figure 3.11, as the players fails to answer the given question on a certain room inside the game, then possible instead of going and starting all over at the beginning of the game like going back to room number 1. Instead, the player will respawn and will be sent from the last checkpoint depending on what room the player have failed to answer a math question and will be served as progress at specific points once they died.



As the player is playing and exploring every room

When its life bar is not enough, the player can increase its life bar

By just picking a bandage that the player can use to increase their life bar

Figure 3.12 Using bandage as a tool to increase life bar

In this figure, when the player is exploring with a low health bar as they explore each area or room, they can look for a bandage that can be used in order for them to increase their life bar and continue throughout the game.

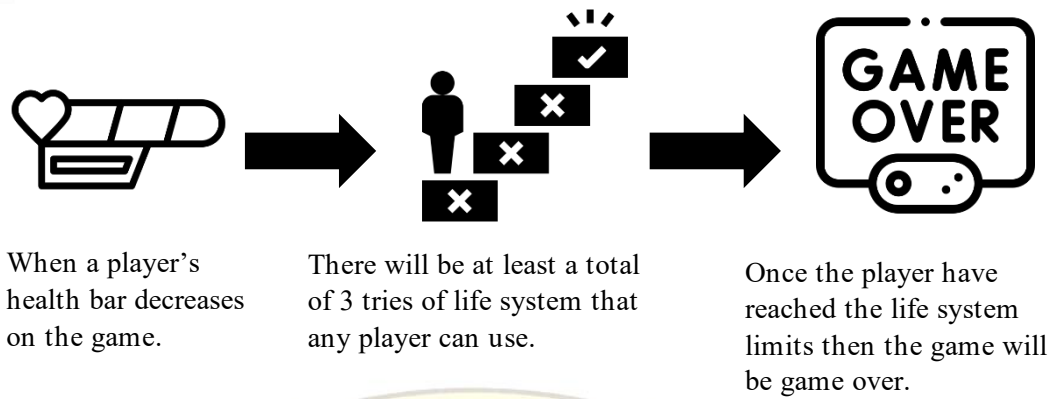


Figure 3.13 Using life system to continue throughout the whole game

In this figure, when the player health bar is almost decreasing and started to go low, there will be 3 tries of life system that they can use throughout the whole game from the beginning until the end but once they have reached the life system limits then it is game over already.

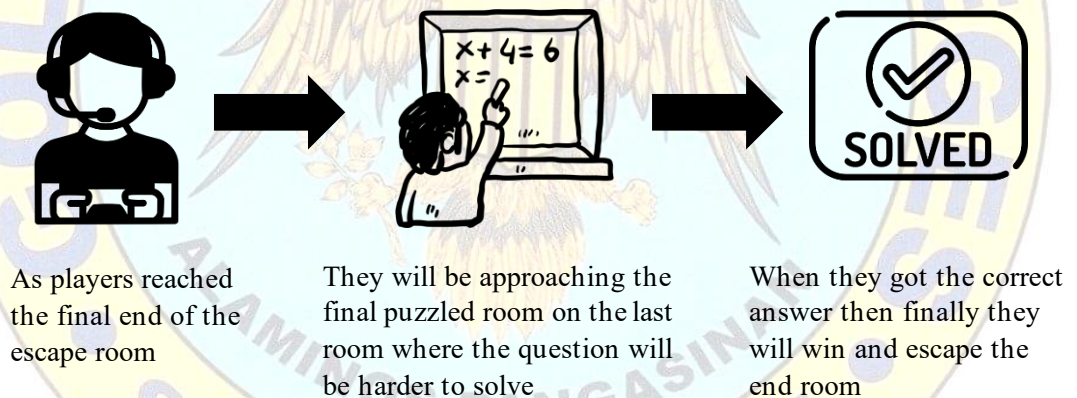


Figure 3.14 Dealing with the final level room at the end of the game

In figure 3.14, as the players have reached the final end of the escape room game play. They will have to face the hardest puzzled room and must solve the question that the question panel will be shown when players interact with it, in order to finish and end the whole game. Players must be keen and fast in order to finally escape the very last room from the escape room game in order to achieve success.

Chapter 4

Discussion of Findings

The Existing Process

The Purposed of this chapter is to present findings of this study. The data collected from the students who have knowledge about the subject mathematics on mobile gaming, specifically a math horror themed escape room where players will be playing after the development of the game. The researchers collected the responses of the 30 students from the College of Information Technology Education Department. Detailed charts accompany each section to illustrate the findings clearly.

The game Escape Room: MathemaTerror procedure will be shown on the following figures below like Figure 4.1, Figure 4.2, Figure 4.3, Figure 4.4, as well as Figure 4.5 wherein this graphical representation will be served as a guide to determine how does the whole game entitled “Escape Room: MathemaTerror” is going to be and how does the gameplay will be once the players from the College of Information Technology Education Department started playing it from the start until the final end whether each gameplay will make players succeed or failed.



Figure 4.1 shows how the players progress through the game.

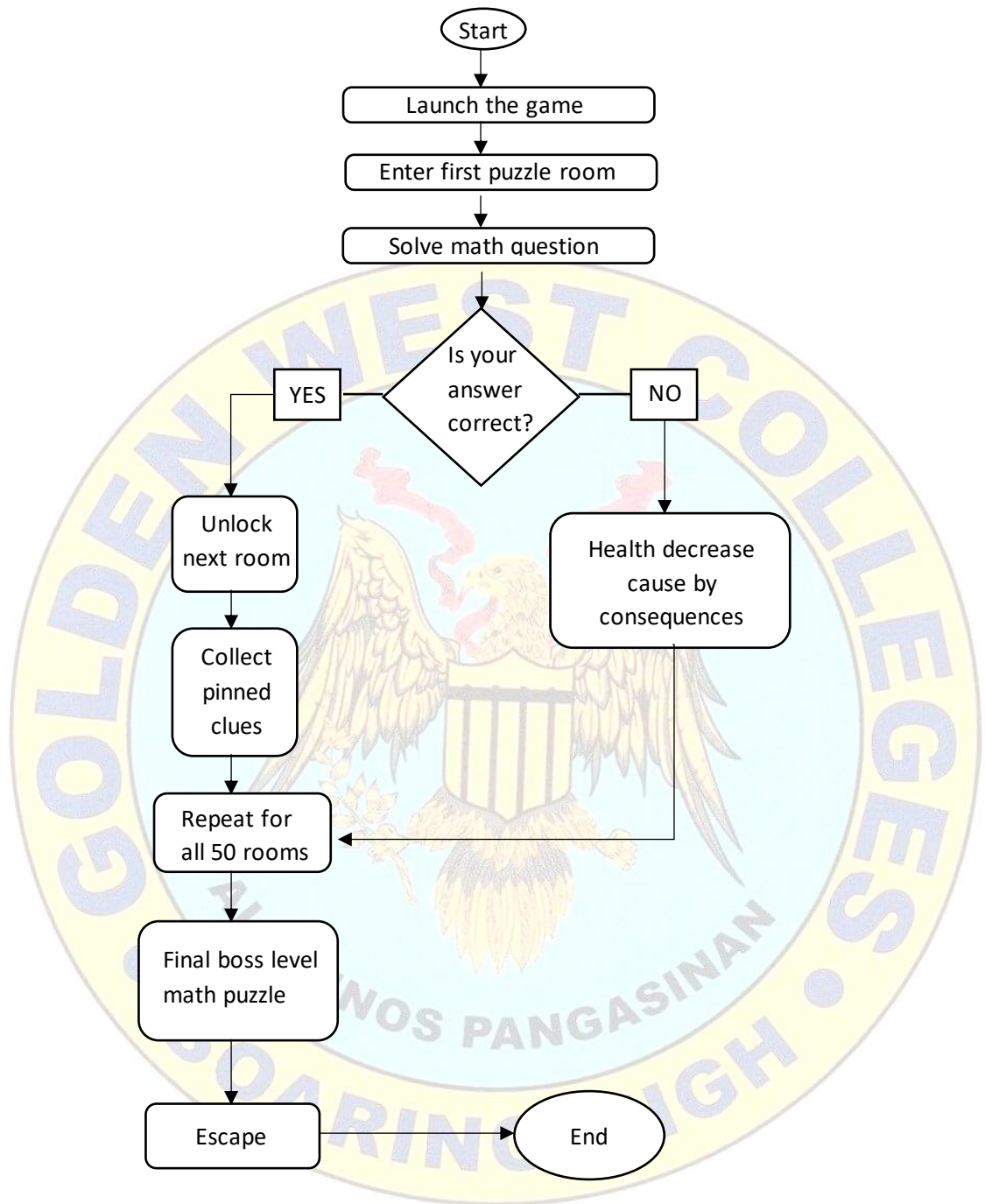


Figure 4.1 Graphical Representation on how the players progress through the game.



The step-by-step process of game play how the players are supposed to proceed through the game, in a suitable and understandable manner as explained is illustrated in the flowchart in Figure 4.1. Whenever the player opens or launches the application and the game begins, this marks the beginning of an official gaming session. Once the player has completed loading, they are taken directly to the first puzzle room, the stage of the level where players are first presented with the mathematical concept. In this room, the player must solve a math problem to gain access to other rooms that has mathematical questions that needs to be answered. As soon as the game chooses its response, it immediately looks to see if the player should have or should not have answered. If the player's answer is already correct, the game allows them to open the next room, unlocking more portions of the game. As every room is unlocked, players can collect clues like a piece of paper that has a pin number on it that are essential to its progress and may lead them to complete other puzzles or solve more complex riddles. Players must repeat this sequence of solving math problems, unlocking new rooms and passing through important objects fifty times.



Figure 4.2 shows how the game flow when a trap is triggered.

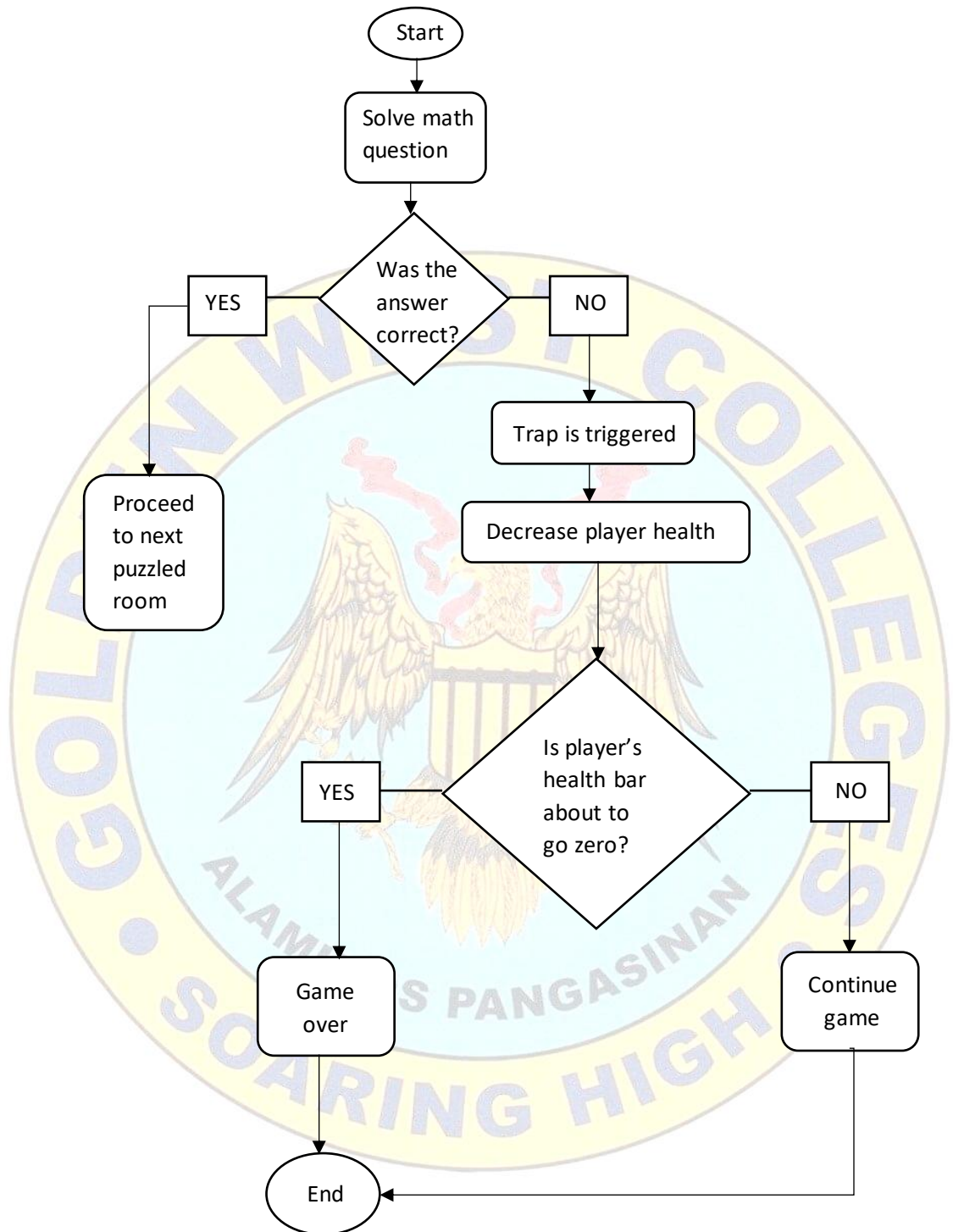


Figure 4.2 Graphical Representation on how the game flow when a trap is triggered.



The game flow that occurs when a trap is triggered in the game is explained in this flow chart which is the Figure 4.2. A math problem is presented to the player at the start of the process. The player will easily continue the game by instantly moving on to the next math puzzled room if they successfully answer the question. However, a trap will be triggered if the player gives the incorrect response. The trap's activation causes the player's health to drop. The game then determines whether the player's health bar is on the edge of going to zero. The game will end and the player will not be able to continue if the health bar is ready to run out. However, the game will continue and the player gets another chance to advance if their health is still not over zero. This flow makes sure that while giving the wrong response results in health loss and the potential to lose the game if health runs out, giving the right answer advances the player. As the player solves the puzzle correctly then it will proceed to the next room until the player reaches to the final end.

Figure 4.3 shows how the players progress through the game with checkpoints.

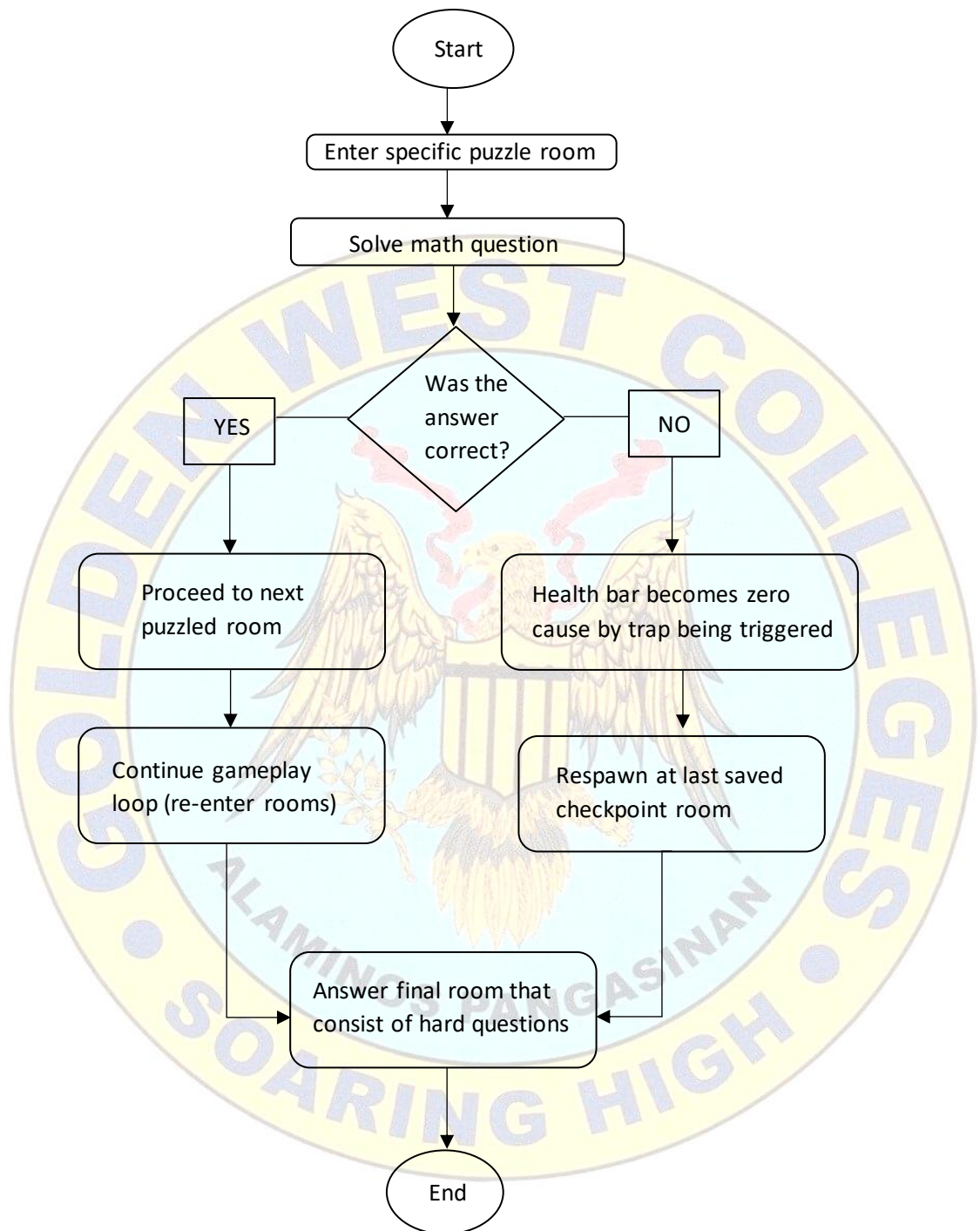


Figure 4.3 Graphical Representation on how the players progress through the game with checkpoints.



The game flow that occurs when a players progress throughout the game with checkpoints in the game is explained in this flow chart which is the Figure 4.3. When a player is answering a question in a specific puzzled room then suddenly the trap triggered afterwards the player's health bar keeps on decreasing until it turns out into zero unless when it is answered correctly then the player will continue the gameplays loop wherein the player's will discover new puzzled room but this is how the checkpoint works: Every five rooms, there is one designated checkpoint room that serves as a respawn point. This means that the player's progress is saved whenever they reach that room, and if they die later on, they will restart from that checkpoint instead of going all the way back to the beginning. When the players reaches a room that has checkpoint on them then the game automatically saves the location as the new checkpoint of the players. The player progresses through rooms in numerical order then lastly from there, the player can continue playing and attempt to pass the remaining rooms again.



Figure 4.4 shows how player can use life system to continue playing the game.

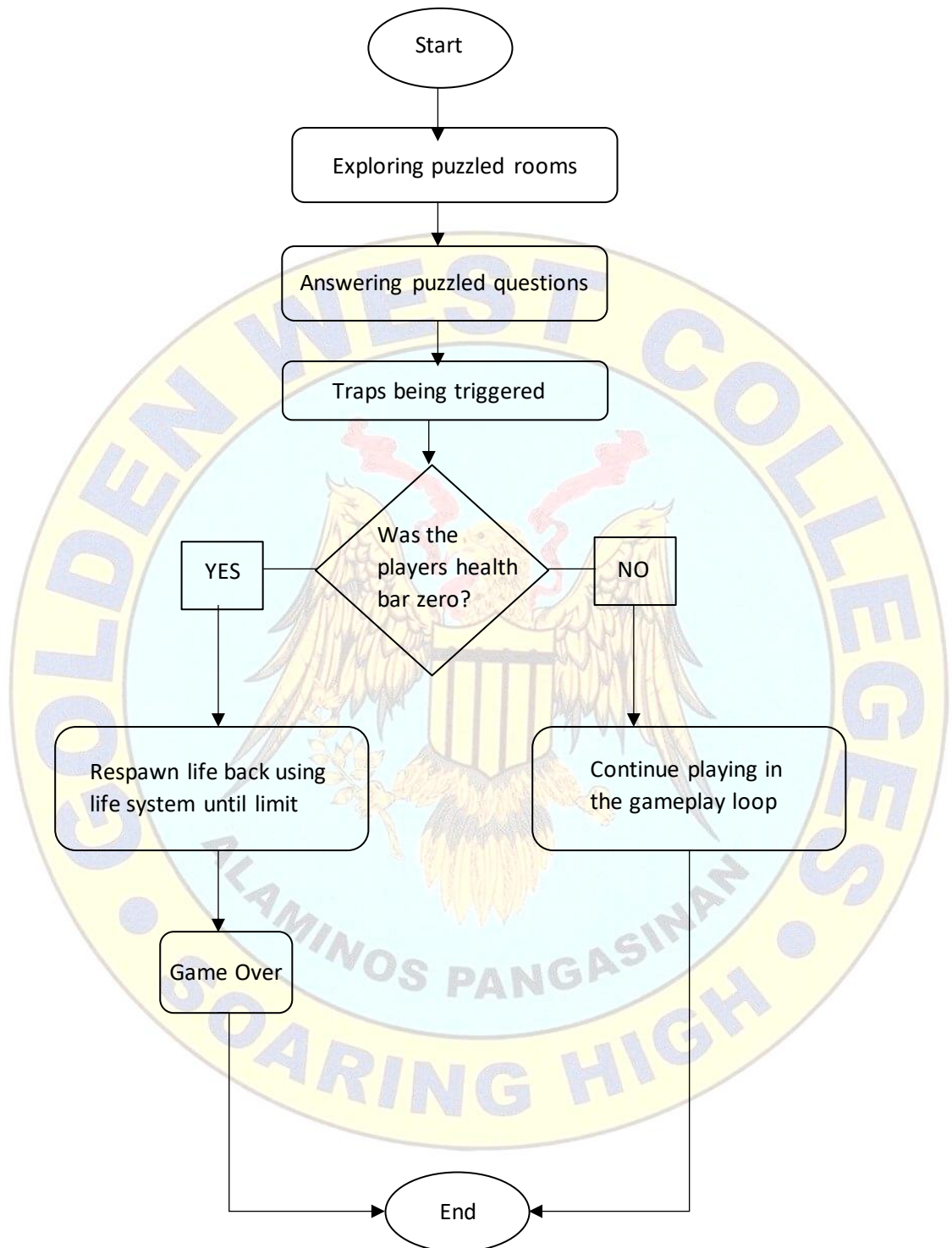


Figure 4.4 Graphical Representation on how player can use life system to continue playing the game.



In this figure, it show that when a player is exploring a puzzled room they need to answer in order to escape and proceed to the other new room that also has different type of question but is all related to mathematics. When the player got the right answer then finally they can go to the next puzzled room but if not and the player did not get the right answer then the trap will be triggered until it decreases its health bar but since the game has the life system, the player can still have some chance to continue playing the game as long as they maintain the life system that only has 3 chances to respawn again in the game but once the 3 life system have reached its limits and turned into zero then the player cannot continue the game since it will be considered as game over. So as much as possible, players must be more careful in maintaining as well as saving the 3 life system chances and lastly must handle the life bar or health bar of any player that is going to play the game entitle “Escape Room: MathemaTerror.”

Figure 4.5 shows how player can increase life bar by using medical kit.

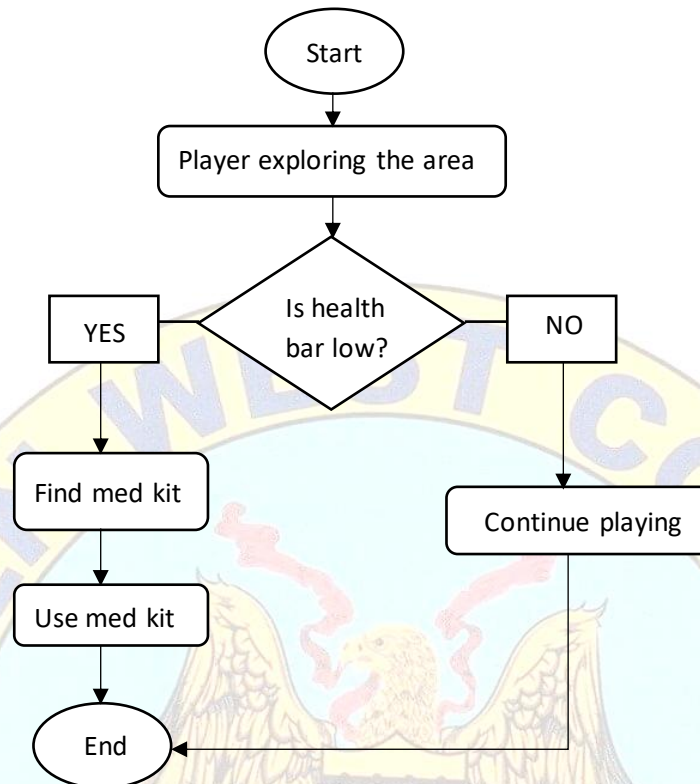


Figure 4.5 Graphical Representation on how players can increase life bar by using bandages.

In this figure, it show that when a player is exploring an area or a room throughout the game. The player can discover a medical kit wherein it can be used when its life or health bar is not enough. Once the player interacts with the medical kit this can increase its life or health bar in order to continue playing and exploring the next room with questions that the players must answer so that it can reach at the final boss and successfully end the game.



Problems Encountered

When dealing with the subject mathematics, students at school tend to feel slow and not knowledgeable because of the complexity of the subject. The developers have reflected on their own experiences towards how challenging the situation is in learning about mathematics.

First, students find the subject mathematics boring and unimportant to their lives, leading to disinterest as well as lack of effort in learning the subject, like simply reading it in textbooks or listening to their instructors in order to understand it in a proper way.

Second, students tend to develop a fear towards math subjects due to past failures or high-pressure environments like maybe getting the answer wrong or being compared to others since they lack skills that causes them to find math difficult.

Third, students another problem with the subject mathematics is that they have poor foundation of basic math skills since of the overreliance in using the calculator most of the time as well as lack of practice and strengthening in understanding math subject that cause students to struggle and deal with advanced math topics.

In developing the game, we tend to make solutions for the problems encountered in current process of students that are having a hard time learning and strengthening its skills in the mathematics subject in order to enhance their logical, analytical and mathematical skills.



Escape Room: MathemaTerror

System Design. With the challenges identified in the previous sections of this chapter, the developers were encouraged to create an innovative solution that would address the learning difficulties faced by students, particularly in the subject of Mathematics. To respond to these issues, the team decided to design and develop a game-based learning computer game which is an interactive mathematics escape-room game entitled Escape Room: MathemaTerror. The main goal of this project is to transform the traditional way of learning mathematics into a more engaging, enjoyable, and interactive experience for students.

This game aims not only to make learning more entertaining but also to enhance the students' problem-solving skills, logical thinking, and mathematical comprehension through the use of challenges and puzzles included within the gameplay. By combining educational content with an exciting horror escape-room theme, students are encouraged to think critically and apply mathematical concepts in order to advance through different stages of the game.

Furthermore, the development of Escape Room: MathemaTerror serves as a modern approach to educational technology, particularly designed for the institution of Golden West Colleges. The use of interactive multimedia, sound effects, and creative design elements allows the application to cater to different learning styles while maintaining the balance between education and entertainment.

Gantt Chart. This represents what each user may interact with within the system and what purpose. Figure 4.6 represents the Gantt Chart of the developed system.

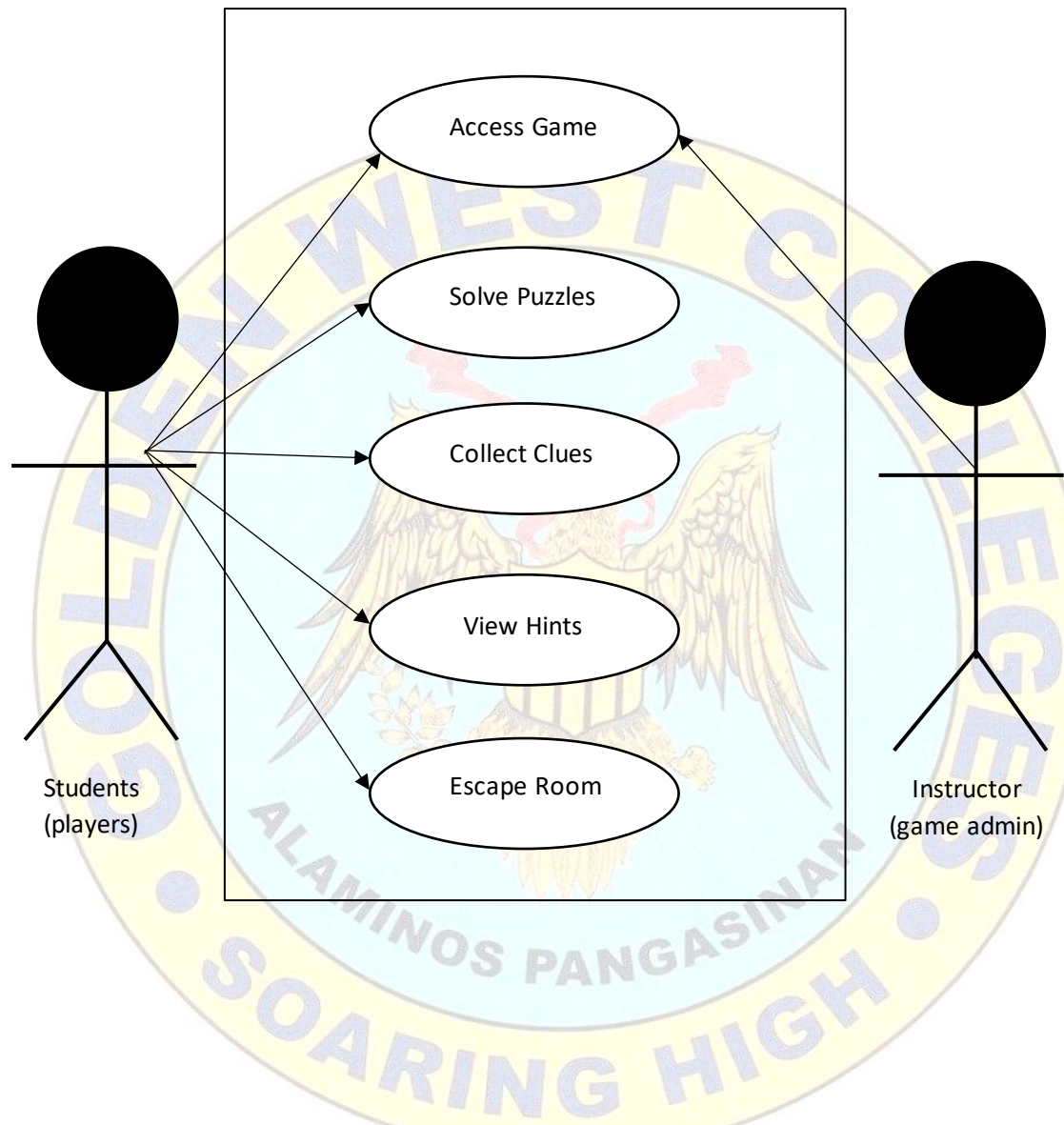


Figure 4.6 Used Case Diagram

The Developed System

To start with the game walkthrough, the Main Menu UI is shown in Plate 4.1



Plate 4.1 Main Menu UI

Main Menu UI. When the player opens the game *Escape Room: MathemaTerror*, the Main Menu User Interface (UI) appears, as shown in Plate 4.1. This is the first screen the player will see when the game starts. It includes buttons such as Play, which starts the game; Settings, and Quit, which closes the game. The main menu is made to be easy to use and understand, helping players quickly start or adjust the game before playing.

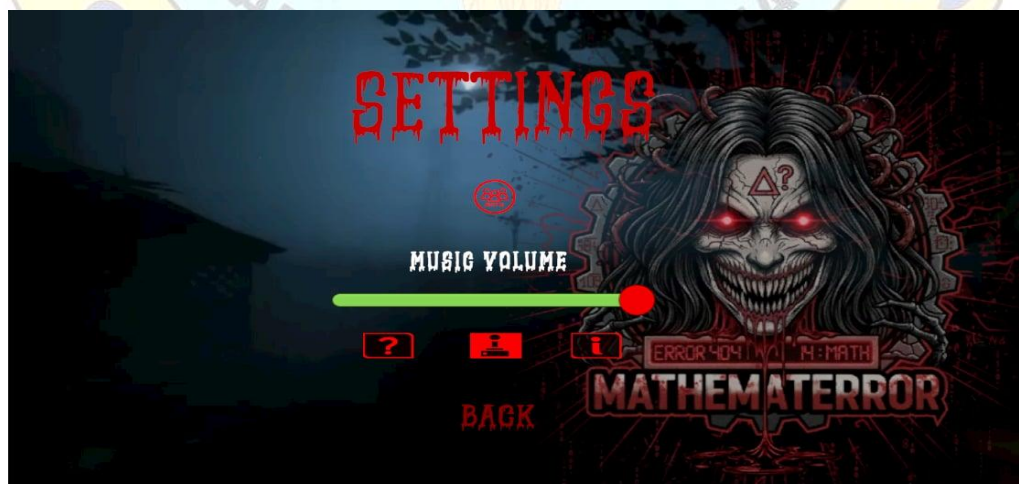


Plate 4.2 Game Setting Screen



Game Setting Screen. This screen shows the Settings section of the game. In this section, there are several buttons such as the About Us button, which shows information about the game developers, the Game Tutorial button, which provides instructions on how to play the game, the Gameplay Controllers button, where players can view or change the game controls; and the Game Information button, which gives more details about the game itself. Players can also adjust the background music volume, from the highest to the lowest level, depending on what they prefer.

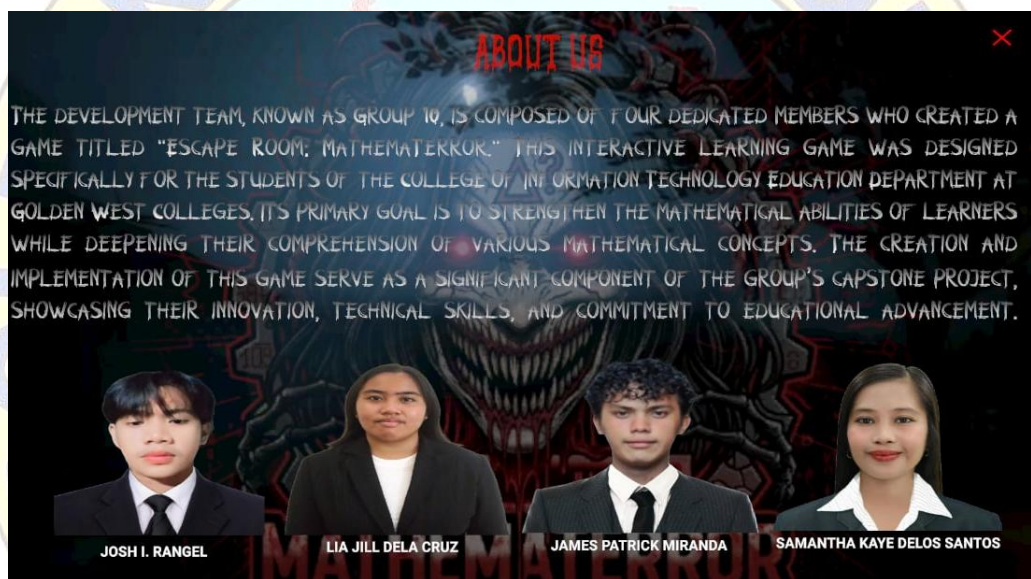


Plate 4.3 Game About Us Screen

Game About Us Screen. This screen shows the basic information about the developers who created the game Escape Room: MathemaTerror. It includes the names of the developers and a short description of their role in making the game. The screen also explains the main purpose of the game, which is to help students learn mathematics in a fun and interactive way. It gives players an idea of why the game was made and what it aims to achieve before they start playing.

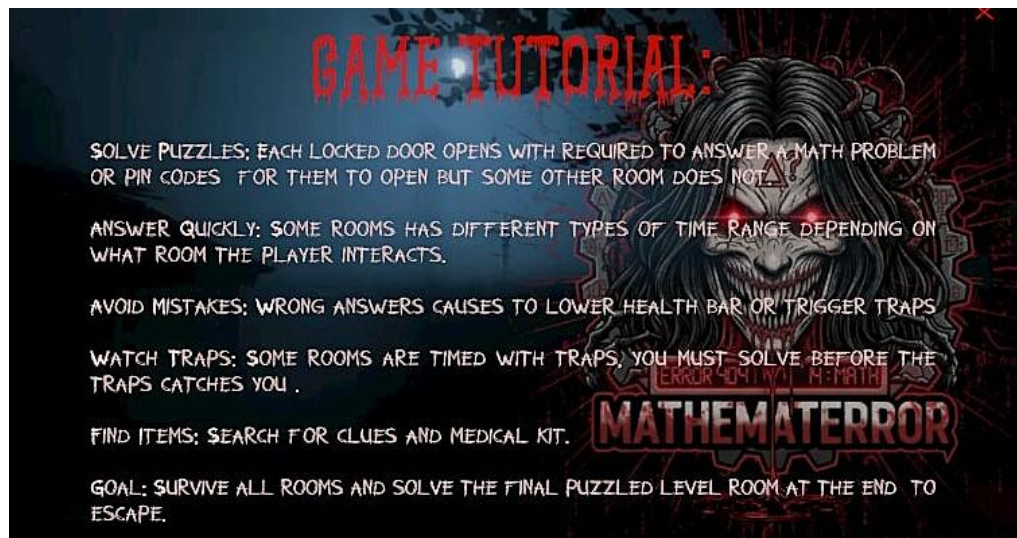


Plate 4.4 Game Tutorial Screen

Game Tutorial Screen. Screen where it provides the step-by-step instructions and guidance on how all players play the game which is the Escape Room: MathemaTerror like the rules, objectives, and lastly the game instructions.



Plate 4.5 Gameplay Controllers

Gameplay Controllers. This screen shows all the keys of the controller that will be used in the game in order to play by its different functions coming from the keyboard as well as the use of a mouse to navigate.

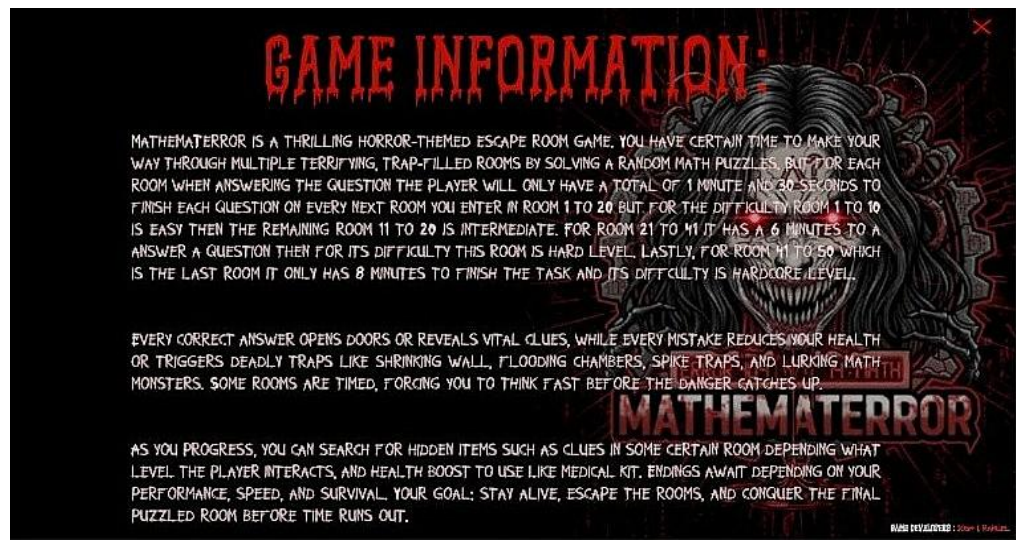


Plate 4.6 Game Information

Game Information. This screen shows the information of the game, Escape Room: MathemaTerror wherein it shows the following: 1. Game Overview, 2. Objectives, 3. Rules and Mechanics, and lastly 4. Progression.



Plate 4.7 Loading Screen

Opened Door. When the players started to click the play button that can be seen in the Main Menu UI before entering the game there will be a loading screen first.

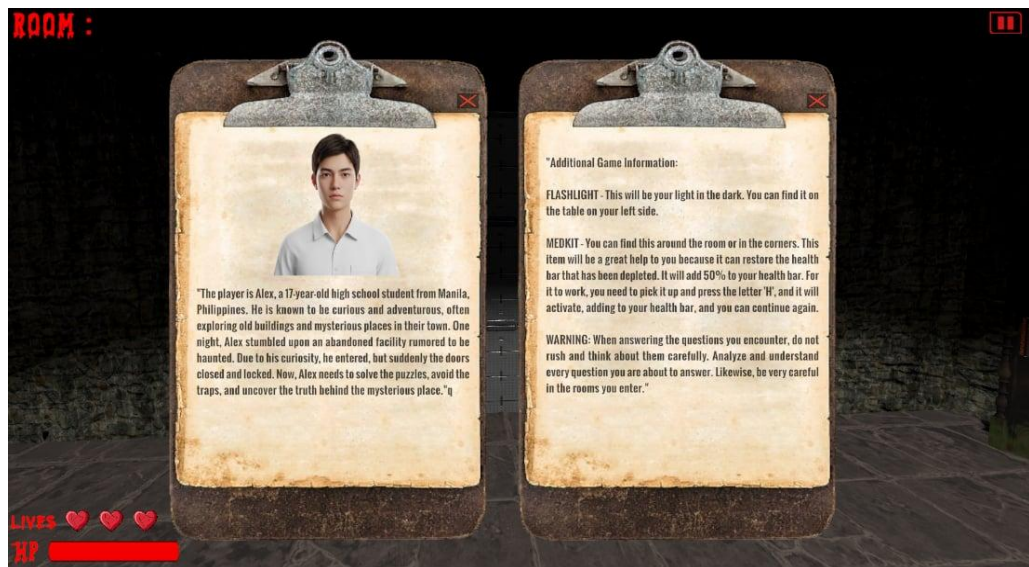


Plate 4.8 Player's Background Story and Basic Gameplay Instructions

Player's Background Story and Basic Gameplay Instructions. When players entered the first ever room, the game will show a different panel where there is background story of the player near the bed and the basic gameplay instructions will be shown near the table before the start of the game.



Plate 4.9 Flashlight as Tool



Flashlight as Tool. As the players begin to explore the area, they will notice a flashlight placed on a table near the room. The flashlight can be picked up and used by the player. Its main purpose is to provide light while exploring dark places in the game. This helps the player see the surroundings more clearly and find hidden objects or clues needed to solve puzzles. The flashlight stays with the player throughout the whole game, making it an important tool for completing tasks and moving forward in the gameplay.

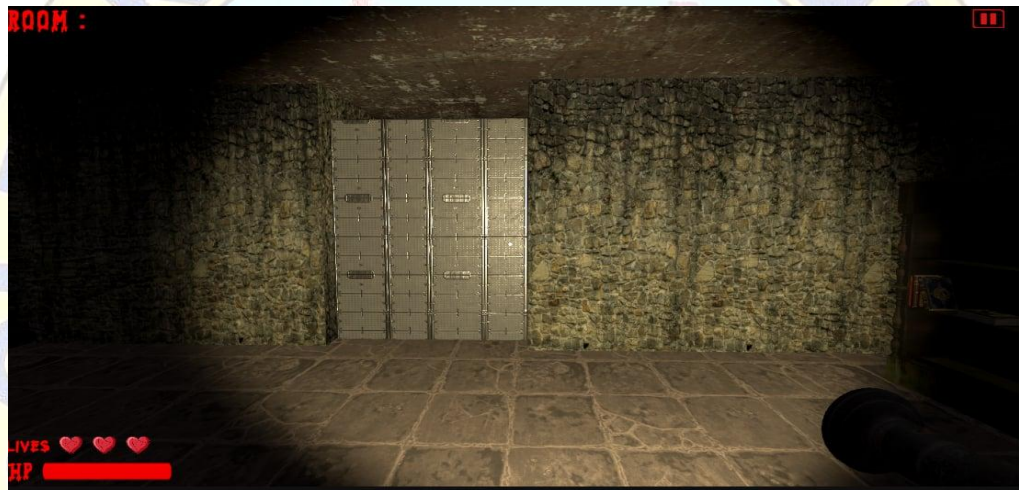


Plate 4.10 Opened Flashlight for Source of Light

Opened Flashlight for Source of Light. The flashlight serves as the main source of light for the player while exploring the game environment. When the player picks up and opens the flashlight, it helps brighten dark areas, allowing them to see objects, clues, and pathways more clearly. It is an important tool in the game because it guides the player through different rooms and helps them avoid missing important details. The flashlight stays on during the whole game, helping the player move around and see clearly even in dark areas.

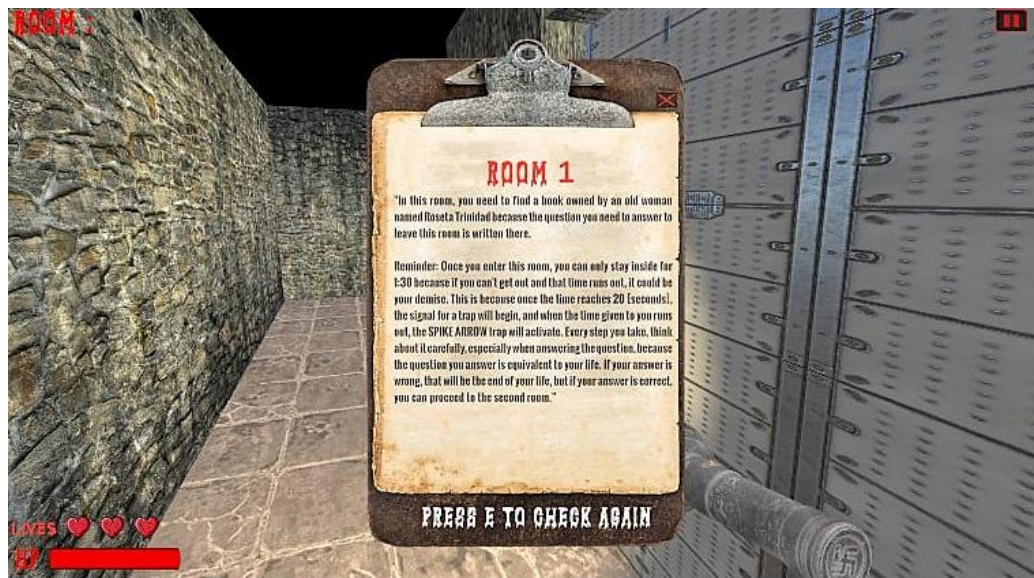


Plate 4.11 Room Information

Room Information. In this plate, before entering a certain puzzled room, there will be a panel that shows the information of a room wherein it talks about what must the player do and avoid once they have entered that room.



Plate 4.12 Timer to Increase Pressure

Timer to Increase Pressure. The timer is used to add challenge to the game. It gives players a limited amount of time to solve puzzles or complete tasks.

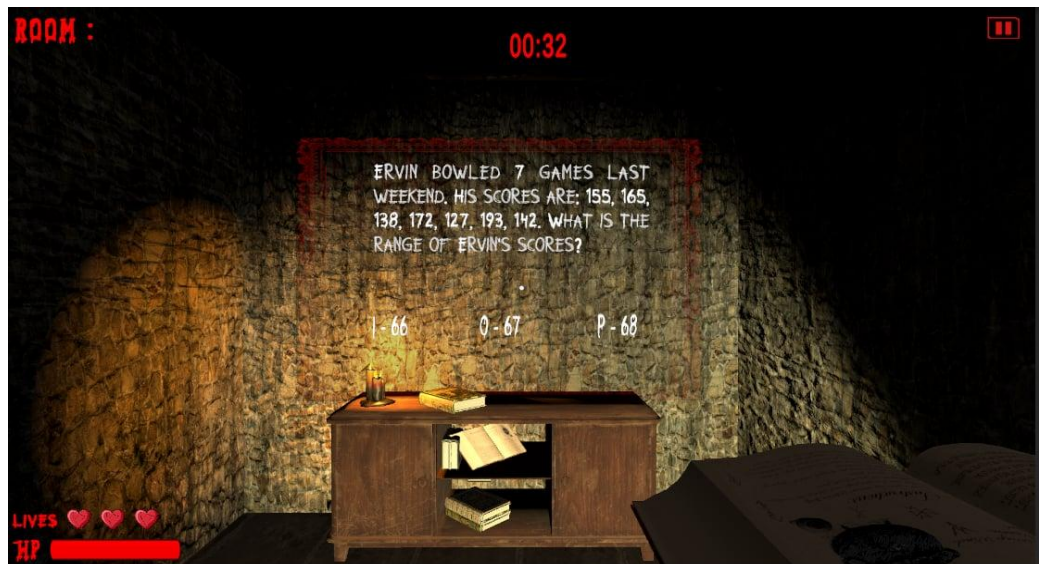


Plate 4.13 Question Panel

Question Panel. In this part of the game, players must explore each room and solve questions to escape. To display the question panel, the player needs to find and interact with a book that triggers it. In some rooms, players will see two boxes and must figure out which one hides the book. Once the book is found, a mathematical question will appear, which the player must answer to progress



Plate 4.14 Jump Scare Panel



Jump Scare Panel. When players explore a certain room or area, they may encounter jump scares that serve as a tool to increase suspense. This adds pressure and excitement while they answer and solve math problems in the game. It also heightens the tension and thrill that players experience throughout the game play until they reach the finale.

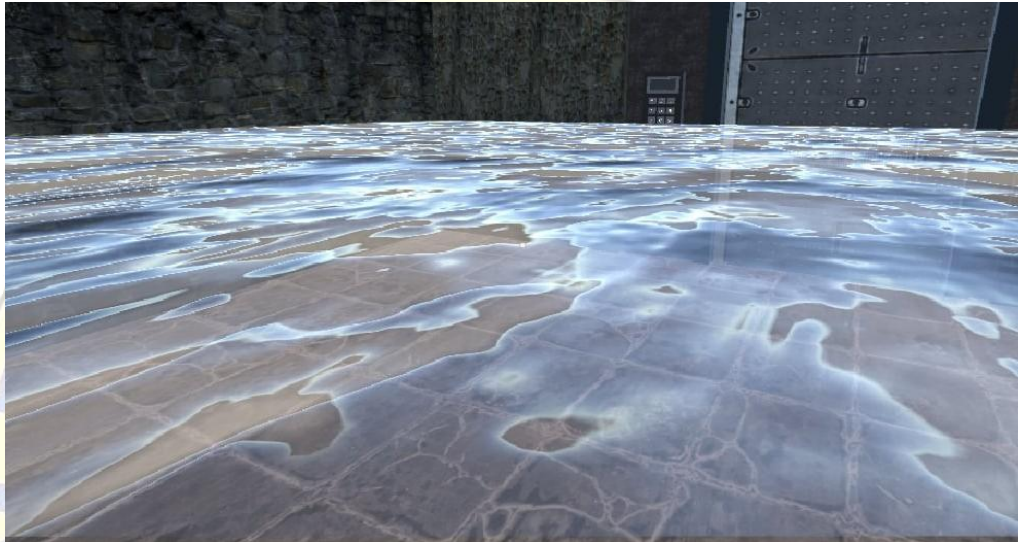


Plate 4.15 Triggered Traps or Hazards

Triggered Traps or Hazards. Based on the plate, the rising water is one of the traps that players may face in the game. It appears when the player fails to solve a puzzle within the given time, making it more challenging and exciting. As the water level rises, the player must hurry to complete the task before it's too late. Aside from the rising water, there are also other traps such as the flying spike trap and the shrinking wall trap, which test the player's focus and quick reaction. These traps are designed to make the game more thrilling and to keep players alert while exploring each room. When a player gets hit or caught by any of these traps, their life bar will decrease, showing that they have taken damage. These hazards remind players to stay alert, think fast, and solve puzzles carefully to avoid losing health and successfully escape each level.



Plate 4.16 Pin Clues for Closed Doors

Pin Clues for Closed Doors. The plate shows that when a player have discovered a clue that has a pin number on it wherein mostly can be seen inside the boxes near the room, players can obtain this once they have interacted with the question panel and got the correct answer on the question that is being shown on a panel. This can also serve as one of the essential gameplay elements that every player can use.

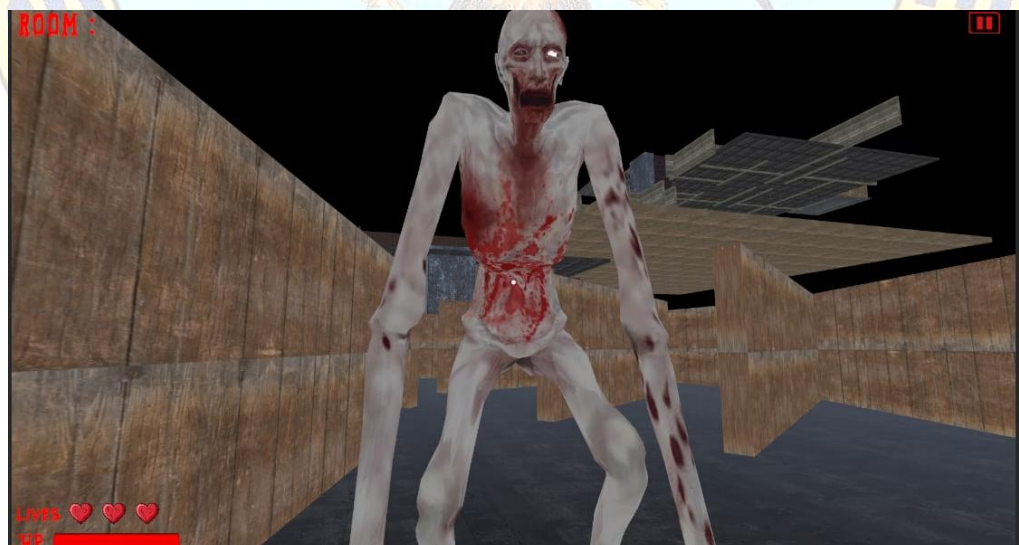


Plate 4.17 Chasing Monster



Chasing Monster. For this plate, the players who is going to play the game will experience a monster that will be chasing them in order to add excitement and thrill while roaming around and on a certain area. Players must be aware of their surroundings so that they can escape and reach the final end of the game.

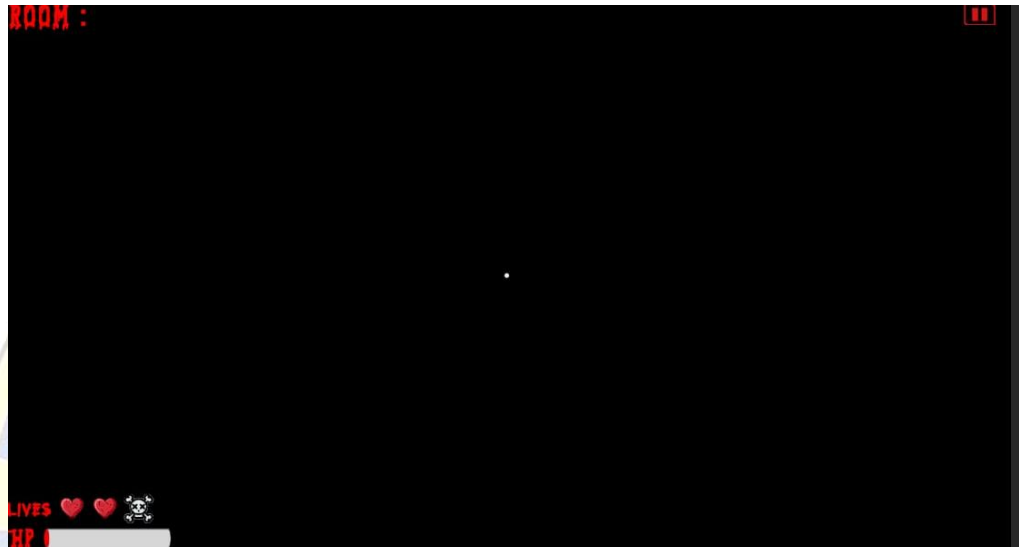


Plate 4.18 Life System

Life System. For this plate, the game uses a three-try life system that limits how many times a player can make mistakes until they die. Each player starts with three lives at the beginning of the game. Every time the player gets caught in a trap, fails a puzzle, or makes a wrong move, one life is taken away depending on the timer and trap they will encounter. When all three lives are lost and the life bar reaches zero, the game will be over, and the player will need to restart the game from the beginning. This system adds challenge and encourages players to be more careful and strategic while playing.



Plate 4.19 Medical Kit for Healing

Medical Kit for Healing. For this plate, the purpose of this medical kit is to restore the players life, which has been drained because of the damages to all traps they have encountered.



Plate 4.20 Checkpoint

Checkpoint. For this plate, once the player have died on room number 4 they will respawns at the room number 1. Checkpoints starts at every 5 rooms, another example if the player dies in room number 9 then they respawns in room 6.

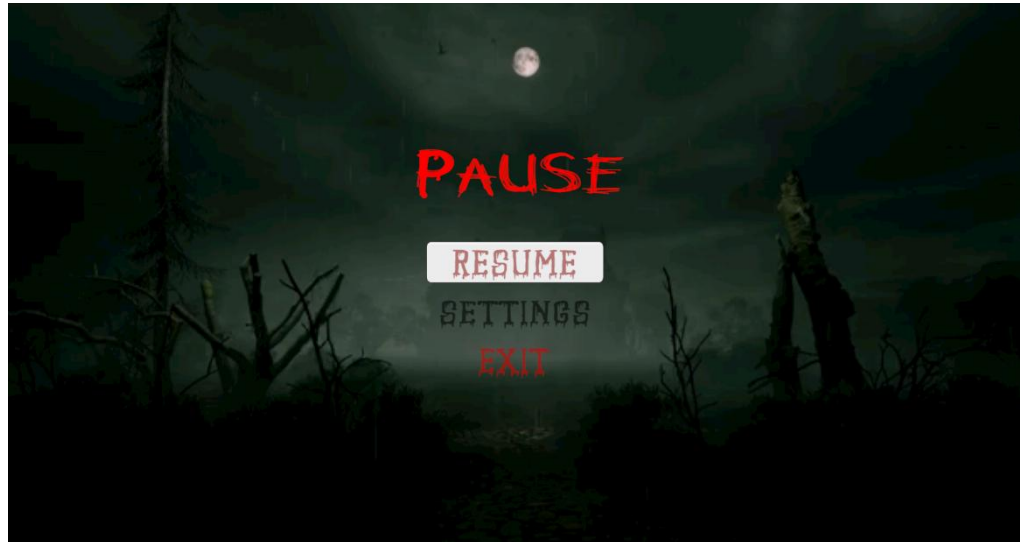


Plate 4.21 Pause Menu UI

Pause Menu UI. For this plate that is being shown is the pause menu UI where the players can locate the resume button, settings button, and lastly the exit button.

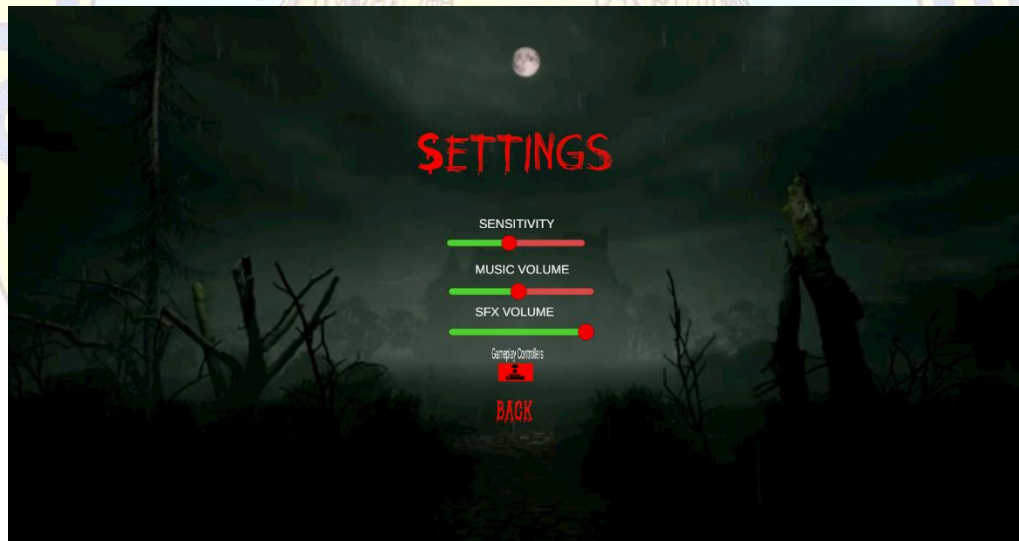


Plate 4.22 Pause Setting UI

Pause Setting UI. Plate 4.22 displays the pause setting user interface. Players can adjust the sensitivity, music volume, and sfx volume by clicking on the setting button. Here in the pause setting user interface, game controls are also visible.



Plate 4.23 Game Over Display Screen

Game Over Display Screen. The Game Over Display Screen appears when the player loses all their lives or fails to complete the objectives. On this screen, the player can easily locate the Restart button to play the game again from the beginning, and the Exit button to leave the game or return to the main menu. It provides a clear and user-friendly way for players to decide whether to try again or quit after a game session.



Survey Results

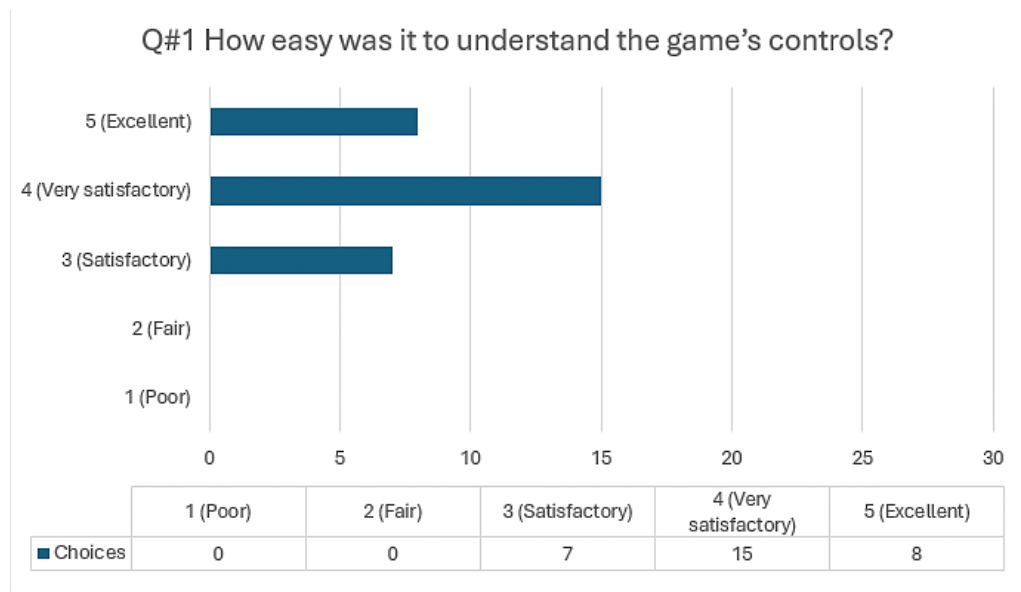


Figure 4.7 Survey Questionnaire #1

The figure 4.7 shows that most of the respondents picked as the highest rating is the choice “4 (Very Satisfactory)” for the first question wherein it talks about if how easy was it to understand the game’s control at the game entitled “Escape Room: MathemaTerror while the lowest one is the “3 (Satisfactory)”.

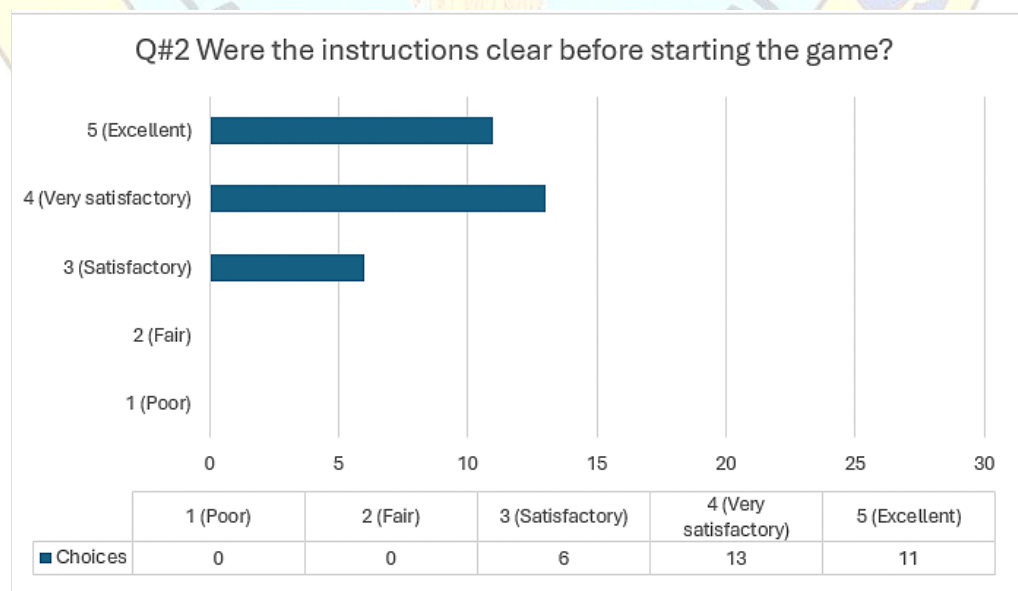


Figure 4.8 Survey Questionnaire #2



The figure 4.8 shows most of the respondent's choose the choice "4 (Very Satisfactory)" on the question were the instructions clear before starting the game, despite the new exposure from the developed game. Overall, the respondents find the instruction on the clear and understandable, While the lowest data from the survey where some respondents choose is the "3 (Satisfactory)".

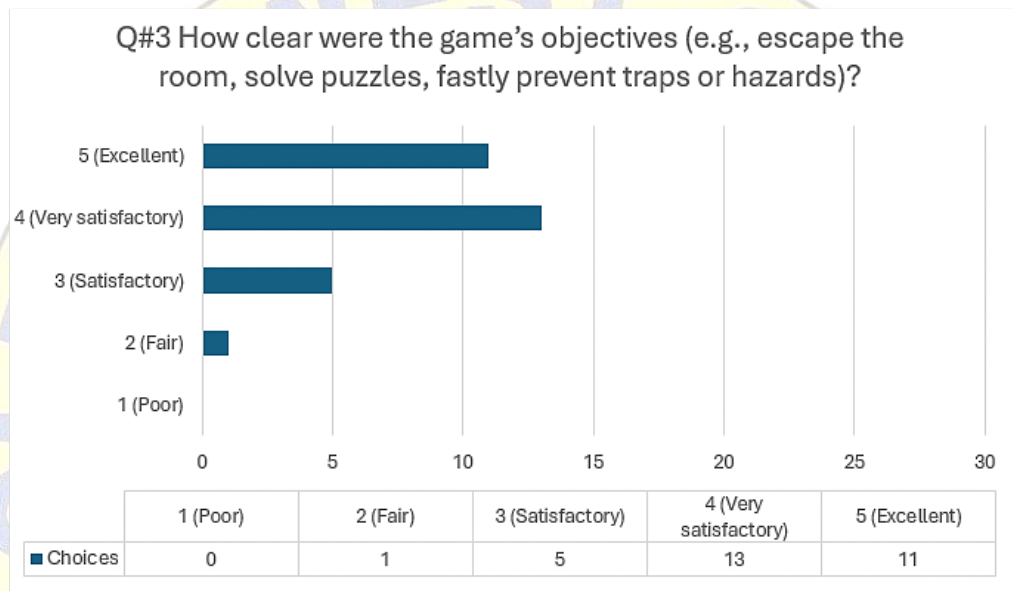


Figure 4.9 Survey Questionnaire #3

The figure 4.9 shows, the highest data that every respondent chose is the "4 (Very Satisfactory)" choice in the question how clear were the game's objectives (e.g., escape the room, solve puzzles, fastly prevent traps or hazards)? While on the other hand, the lowest data in the figure where the respondents least choice is the "2 (Fair)".

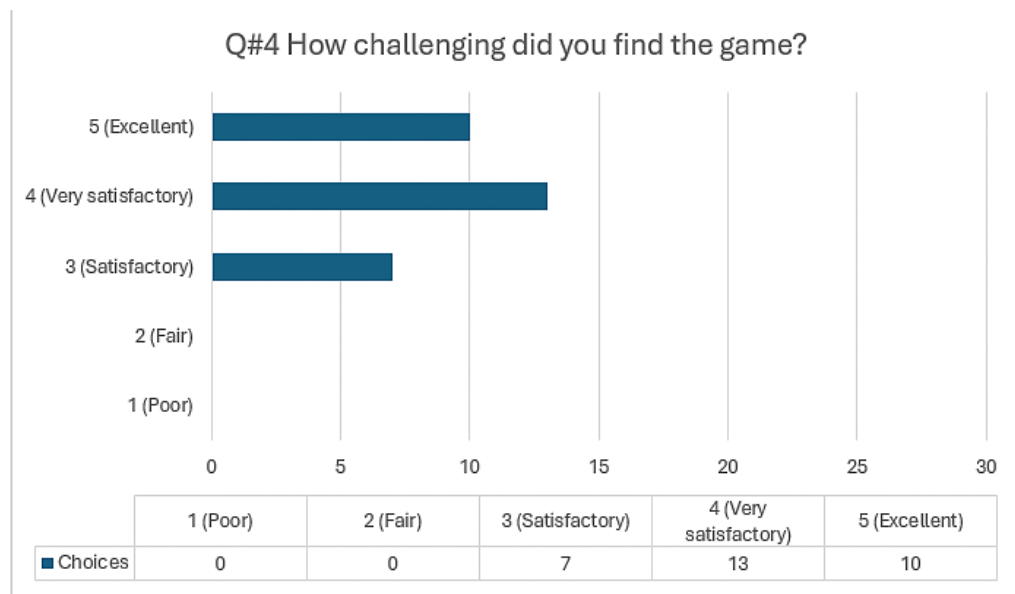


Figure 4.10 Survey Questionnaire #4

The figure 4.10 shows that the highest data gathered from the survey questionnaire #4 is that every respondent chose the “4 (Very Satisfactory)” choice in the game while the lowest data from the figure is the “3 (Satisfactory)” choice.

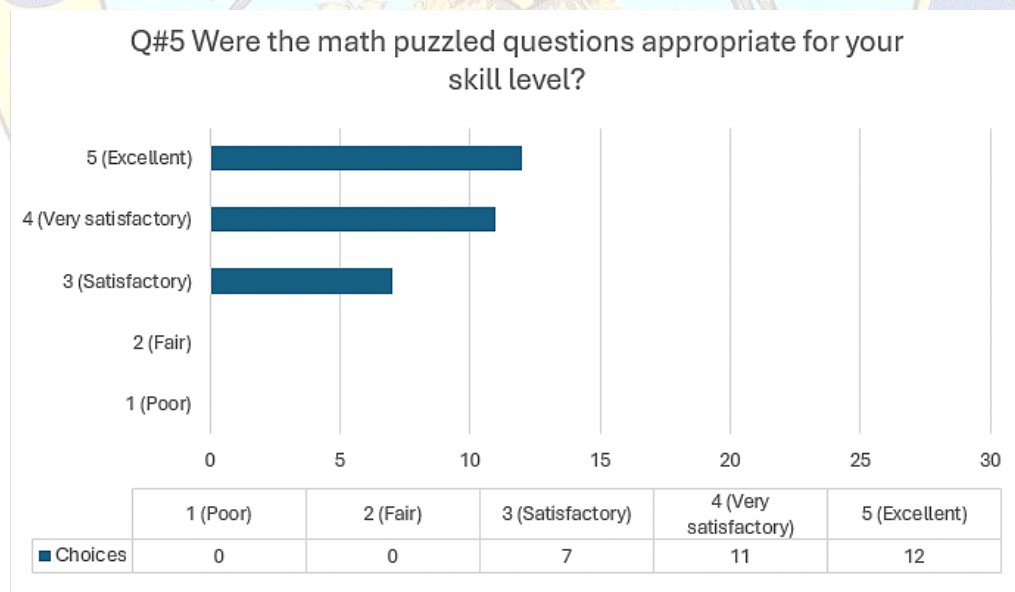


Figure 4.11 Survey Questionnaire #5



The figure 4.11 shows that most of the respondents chose the “5 (Excellent)” at the questionnaire resulting that every math questions in the game is suited for the skills of every respondents who played the game entitled “Escape Room: MathemaTerror” while on the other hand, the most least data from the survey is the “3 (Satisfactory)” choice in the question that tells about if the math puzzled questions appropriate for their skill level.

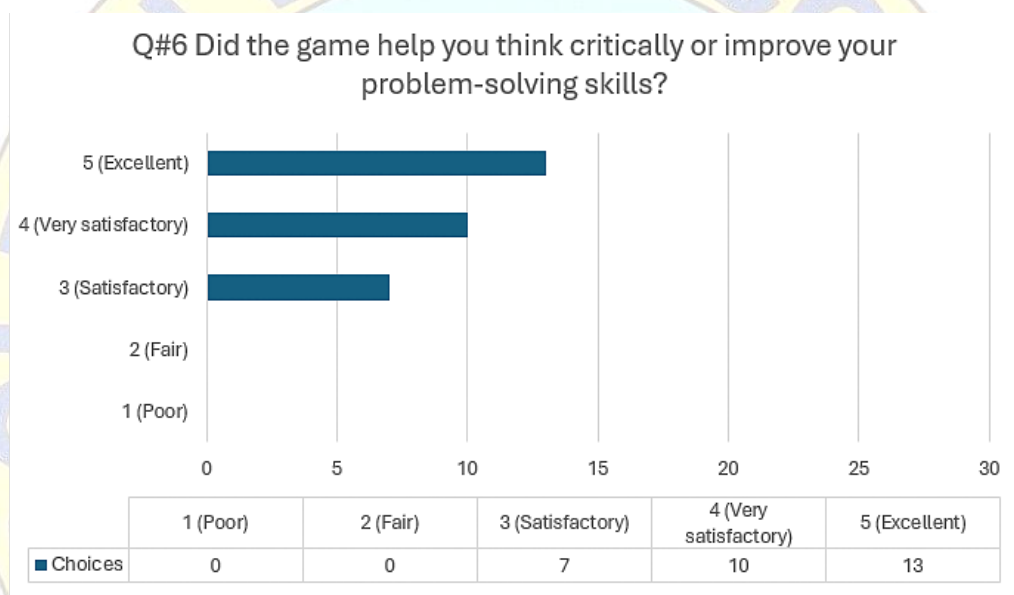


Figure 4.12 Survey Questionnaire #6

The figure 4.12 shows that the highest data of choice that all respondents choose is the “5 (Excellent)” choice in the question about if the game help them to think critically or improve their problem-solving skills while playing the game. While on the other hand, the lowest data of choice is the “3 (Satisfactory)” choice from the diagram.

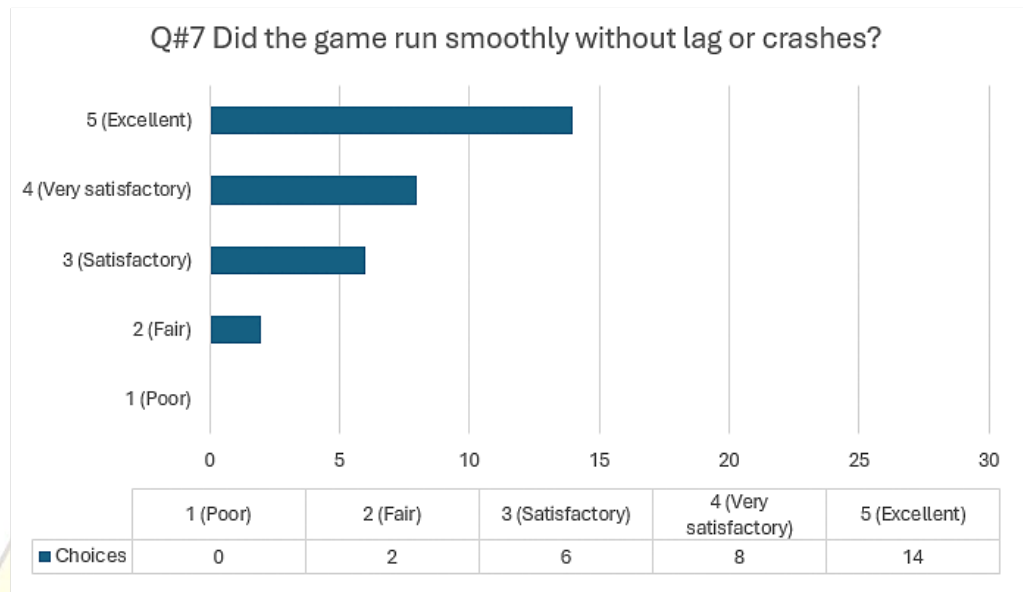


Figure 4.13 Survey Questionnaire #7

The figure 4.13 shows that most of the respondents absolutely think that the horror game runs smoothly without lag or crashes with a total of 14 respondents who picked the “5 (Excellent)” choice while the lowest one with 2 respondents experienced a fair one while playing the game.

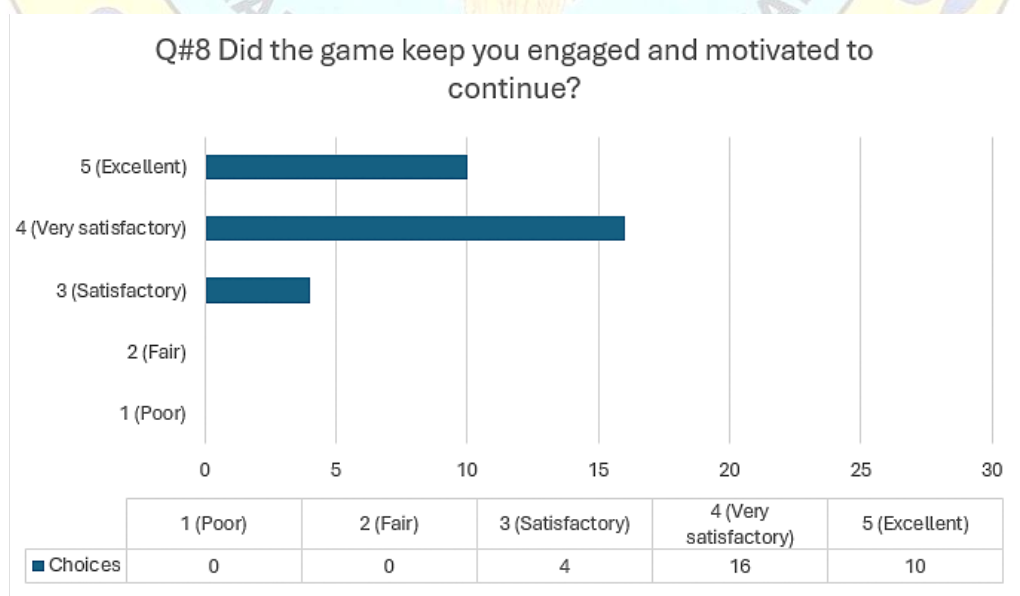




Figure 4.14 Survey Questionnaire #8

The figure 4.12 shows that most of the respondents choose the “4 (Very Satisfactory)” choice on figure while the lowest data from the figure is the “3 (Satisfactory)” choice from the question did the game keep the players engaged and motivated to continue playing the game “Escape Room: MathemaTerror”.

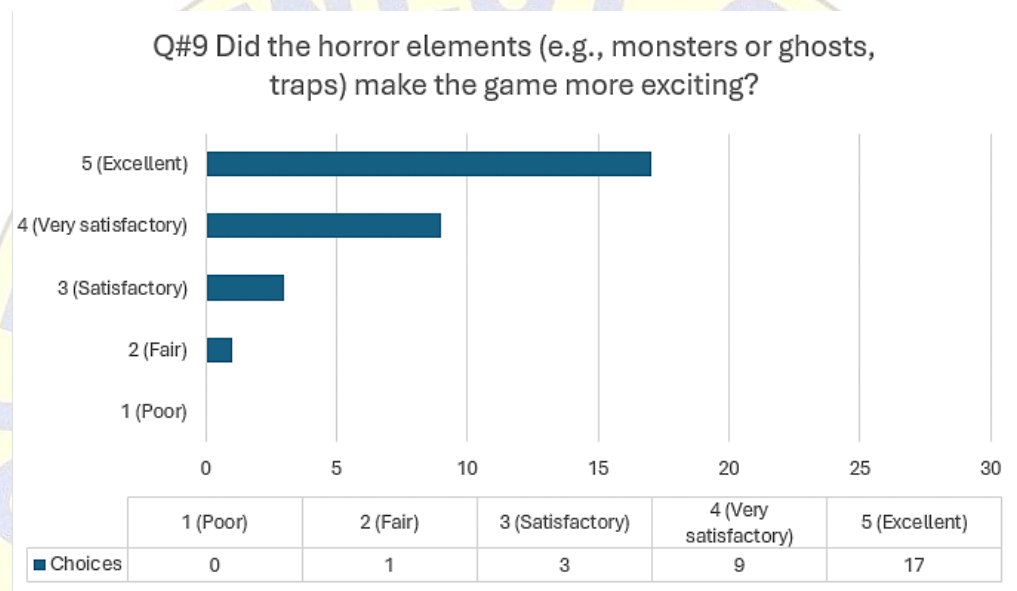


Figure 4.15 Survey Questionnaire #9

This figure 4.15 shows that most of the respondent’s picks the “5 (Excellent)” choice as the answer for the question if did the horror elements like monsters or ghosts as well as traps make the game more exciting while interacting with the game? While on the other hand, the lowest result of data from the figure is the “2 (Fair)” choice from the diagram that showing the results from the usability test question.

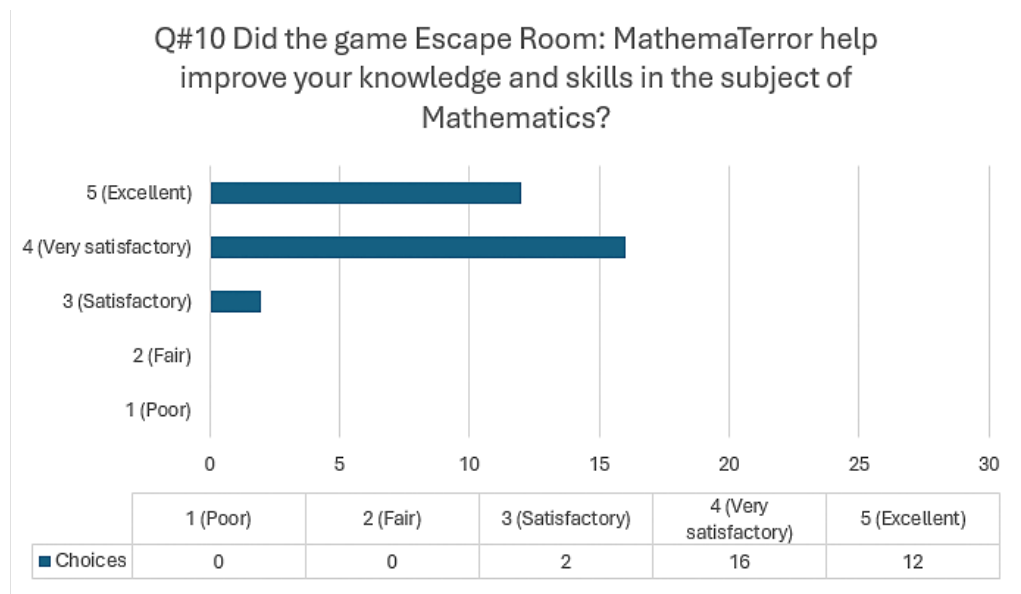


Figure 4.16 Survey Questionnaire #10

This figure 4.16 shows that most of the respondents think that the game “Escape Room: MathemaTerror” improves their knowledge and skills in the subject of Mathematics. The most highest result from the question is the “4 (Very Satisfactory)” choice which is an indicator that the development of the game can be served as a tool that every respondent from the College of Information Technology Education department whose been testing the game during the process of the usability test that the developers have implemented inside the institution of the Golden West Colleges. On the other hand, the lowest result from the question is that some respondent think that it is the choice “3 (Satisfactory)”.

Chapter 5

Summary, Conclusions and Recommendations

Summary

The proposed system begins with a statement of problem which identifies the challenges from the existing system. Objective and Research Questions for the study were stated to direct the emphasis of the work.

The primary objective of the study is to develop a game-based learning application for the College of Computer Science and Information Technology department at Golden West Colleges.

To attain primary objective of this study, the developers specially aim to accomplish that the developed game would help the students at the College of Information Technology Education Department at the Golden West Colleges specifically who struggles in learning mathematics and their expectations while playing the game, Escape Room: MathemaTerror is a learning game-based application, a horror themed game that can captive students' attention, making math concepts more relatable and enjoyable wherein it would enhance their critical thinking and problem-solving skills and learn in understanding the mathematics subject.



Rapid Application Development (RAD) is a software development technique that was used to design and develop the system. The developers used RAD to meet the objectives.

In Golden West Colleges, the developers conducted a usability test with selected students from the College of Information Technology Education Department to evaluate the effectiveness and user-friendliness of the developed game. The usability test served as an essential part of the development process, allowing the team to observe how users interacted with the game and to identify any areas that required improvement.

During the testing phase, the developers prepared a structured testing procedure that guided participants through various stages of gameplay. After interacting with the game, they were encouraged to share their feedback and observations regarding the game's functionality, interface design, level of difficulty, and overall user experience. The information gathered from the usability test provided valuable insights into how users recognized and experienced the game. This helped the developers determine whether the game successfully achieved its educational goals



Conclusion

With the details laid out and the findings about the status of the existing game explained, the followings are the conclusions of this study:

1. Escape Room: MathemaTerror for the students at College of Information Technology Education Department inside Golden West Colleges will be used as a game-based learning tool in order to gain logical, analytical, as well as problem-solving in the Mathematics subject that is being instructed by an instructor inside the institution of Golden West Colleges.
2. The problem encountered from the existing data that has been gathered with the developed game entitled "Escape Room: MathemaTerror is appropriately considered and will be used that can keep students motivated while reinforcing essential skills at Mathematics subject.
3. The features of the game, Escape Room: MathemaTerror has been identified and would be optimized to help aid in preventing difficulties for the students studying related to the Mathematics subject that is being taught on the College of Information Technology Education Department; and
4. The developed game will provide a new platform for the department of College of Information Technology Education Department that can modernize its teaching method while using a game-based learning tool like the developed game, support institutional goals for the department, and lastly promote deep learning that reduces learning barriers while mastering Mathematics.



Recommendations

To enhance gameplay experience, balance difficulty, and ensure smoother performance, the following recommendations are proposed for future development:

1. **Adjustable Time Settings.** Implementing adjustable time limits per difficulty level is recommended to suit different user skill levels.
2. **Puzzle Randomization System.** Randomly generate math puzzles each time a player replays a room, preventing memorization and encouraging critical thinking.
3. **Hint System or AI Assistant.** Introducing a limited use hint system or AI guide is recommended to assist players while maintaining game balance and balance.
4. **Voice Interaction or Narration System.** Add narration or voice hints for accessibility and engagement, especially for visually inclined learners.
5. **Mini-Games and Bonus Rounds.** Introduce short math challenges or quizzes between rooms to break routine and reward quick learners.
6. **Rewards and Unlockables.** Players could unlock new skins, room designs, or horror effects after completing levels or achieving high accuracy.
7. **Storyline Expansion & Chapters.** Add new chapters or expansion packs with different horror settings like haunted school, laboratory, or library to sustain player interest.



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APPENDIX A

Gantt Chart





APPENDIX B

Usability Test Survey Questionnaires

Legend:

5: Excellent, 4: Very Satisfactory, 3: Satisfactory, 2: Fair, 1: Poor

USABILITY	1	2	3	4	5
How easy was it to understand the game's controls?					
Were the instructions clear before starting the game?					
How clear were the game's objectives (e.g., escape the room, solve puzzles, fastly prevent traps or hazards)?					
How challenging did you find the game?					
Were the math puzzled questions appropriate for your skill level?					
Did the game help you think critically or improve your problem-solving skills?					
Did the game run smoothly without lag or crashes?					
Did the game keep you engaged and motivated to continue?					
Did the horror elements (e.g., monsters or ghosts, traps) make the game more exciting?					
Did the game <i>Escape Room: MathemaTerror</i> help improve your knowledge and skills in the subject of Mathematics?					

Course: _____

Year level and set: _____

Signature over the Printed Name



APPENDIX C

Sample Survey

19



CAPSTONE



System: **ESCAPE ROOM: MATHEMATERROR**

4th YEAR BSIT

Overview: *MathemaTerror* is a horror-themed, single-player escape room game for students with a math subject where players must solve challenging math problems to survive. The game combines the thrill of puzzle-solving with the tension of survival horror, placing players in a series of terrifying, trap-filled rooms. To escape, the player must answer **math questions** within a time limit correctly.

Survey Evaluation Form for the Proposed System

Legend:

5: Excellent, 4: Very Satisfactory, 3: Satisfactory, 2: Fair, 1: Poor

USABILITY	1	2	3	4	5
How easy was it to understand the game's controls?				/	
Were the instructions clear before starting the game?					/
How clear were the game's objectives (e.g., escape the room, solve puzzles, fastly prevent traps or hazards)?				/	
How challenging did you find the game?					/
Were the math puzzled questions appropriate for your skill level?					/
Did the game help you think critically or improve your problem-solving skills?					/
Did the game run smoothly without lag or crashes?					/
Did the game keep you engaged and motivated to continue?					/
Did the horror elements (e.g., monsters or ghosts, traps) make the game more exciting?					/
Did the game <i>Escape Room: MathemaTerror</i> help improve your knowledge and skills in the subject of Mathematics?					/

Course: BSIT

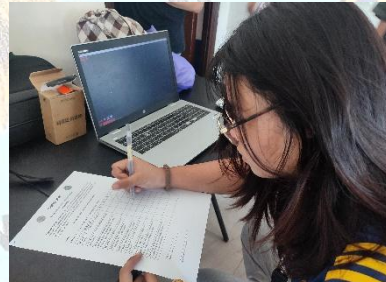
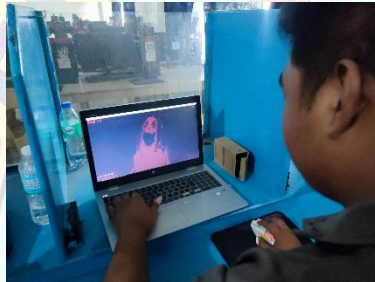
Year level and set: 3 A

DAN BERNARD DE LEON
Signature over the Printed Name



APPENDIX D

Survey Documentation





APPENDIX E

User Manual

Using UNITY for accessing a Unity Project:

Step 1: Install Unity Hub and Unity Editor

1. Download and install Unity Hub from <https://unity.com/download>.
2. Open Unity Hub.
3. Install the required Unity Editor version that matches the project.
 - You can check the correct version in the project's file:
 - Navigate to ProjectFolder/ProjectSettings/ProjectVersion.txt.
 - In Unity Hub, click Installs > Add to install that version.

Step 2: Add the Project to Unity Hub

- Open Unity Hub.
- Go to the Projects tab.
- Click the Add button.
- Browse to the folder where the Unity project is located.
- Select the project folder and click Open.
- The project will now appear in your Unity Hub project list.

Step 3: Open the Project

- In Unity Hub, find the project in the Projects list.
- Make sure it uses the correct Unity version (shown next to the project).
- Click on the project name to open it in the Unity Editor.
- Wait for Unity to load and import all assets (this may take a few minutes).



Step 4: Open the Main Scene

- In the Unity Editor, open the Project window (bottom panel by default).
- Navigate to the Assets/Scenes folder (or the folder specified in project instructions).
- Double-click the main scene file (e.g., MainScene.unity or SampleScene.unity).
- The scene will load in the Hierarchy and Scene View.

Step 5: Run the Project in Play Mode

- Ensure the correct scene is open.
- Click the Play button (triangle icon at the top center of the Unity Editor).
- Interact with the project as intended.
- To stop, click the Play button again

Step 6: Verify Project is Working

- If successful, the project will run inside the Editor without errors.
- Check the **Console** window for warnings or errors.
- Confirm that the expected game objects, UI, or experiments appear in the Scene.



CURRICULUM VITAE



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Tagalog

About Me

A creative Unity game developer skilled in designing and programming engaging gameplay and interactive puzzles. Passionate about building immersive player experiences with strong problem-solving and teamwork abilities.

Education

● 2010-2016

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● 2016-2022

Alaminos City, Pangasinan

Junior-Senior High School

Alaminos City National High School

References

Kairos Jil I. Molina

Instructor, College of Computer
Science and Information Technology
Golden West College's



LIA JILL DELA CUESTA

QA ANALYST

Contact

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Personal details

Age: 22 years old
Gender: Female
Birthday: May 24, 2003
Religion: Roman Catholic

Expertise

- Web designing
- Graphic designing
- SketchUp
- Canva
- Video Editing

Language

English
Tagalog

About Me

A detail-oriented QA analyst with experience in software testing, bug tracking, and ensuring product quality through manual and automated testing. Skilled in test case design, defect reporting, and collaborating with develop.

Education

- 2010-2013
Mandaluyong, Philippines
Elementary School
Famous Academy School
- 2013-2016
Burgos, Pangasinan
Elementary School
Burgos Central School
- 2016-2022
Burgos, Pangasinan
Junior-Senior High School
Burgos National High School

References

Kairos Jil I. Molina

Instructor, College of Computer
Science and Information Technology
Golden West College's



SAMANTHA KAYE DELOS SANTOS

IT SUPPORT ENGINEER

Contact

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Pangasinan

Personal details

Age: 22 years old
Gender: Female
Birthday: March 18, 2003
Religion: Catholic

Expertise

- Web designing
- Graphic designing
- SketchUp
- Canva

Language

English
Tagalog

About Me

An It support with experience in troubleshooting hardware/software issues, providing technical assistance, and ensuring smooth system operations. Skilled in problem-solving, and customer support,

Education

- 2010-2015
Alaminos City, Pangasinan
Elementary School
Binerangan Elementary School
- 2015-2021
Alaminos City, Pangasinan
Junior-Senior High School
Binerangan National High School

References

Kairos Jil I. Molina
Instructor, College of Computer
Science and Information Technology
Golden West College's



JAMES PATRIC MIRANDA

WEB DESIGNER

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Brgy. Mona Alaminos City,
Pangasinan

Personal details

Age: 22 years old
Gender: Male
Birthday: January 23, 2003
Religion: Iglesia ni Cristo

Expertise

- Web designing
- Graphic designing
- SketchUp
- Canva

Language

English
Tagalog

About Me

An innovative web designer who is willing to learn in order to give insights and ideas that could help your company through my hard work and dedication for the job.

Education

- 2010-2015
Alaminos City, Pangasinan
Elementary School
Pangapisan Integrated School
- 2015-2021
Alaminos City, Pangasinan
Junior-Senior High School
Alaminos City National High School

References

Kairos Jil I. Molina

Instructor, College of Computer
Science and Information Technology
Golden West College's